Kidney & UTUC

Condensed rapid revision · facts already answered in Papers 1–4 point to that answer

highlighted number to memorise · **KEY POINT** and **ALGORITHM** boxes carry the reasoning · MCQ exam trivia · † added beyond the source notes · P2 topic = covered in full in that Paper's answer book

Chapter	Topic
Ch 136	Benign Renal Tumours
Ch 137	Malignant Renal Tumours
Ch 138	Urothelial Tumours of the Upper Tract & Ureter
Ch 139	Surgical Management of UTUC
Ch 141	Contemporary Open Surgery of the Kidney
Ch 142	Laparoscopic & Robotic Surgery of the Kidney (incl. donor nephrectomy)
Ch 143	Focal Therapy for Renal Tumours
Ch 144	Treatment of Advanced RCC
★	Putting it together

136 Benign Renal Tumours

1 · The benign small renal mass

Why this matters: many resected "cancers" are benign.

- Up to **~25%** of surgically treated small renal masses (SRMs) are benign.
- Only the **simple cyst** and **fat-containing angiomyolipoma (AML)** are imaging diagnoses. Size, sex and age are unreliable.
- Management follows the risk of renal cell carcinoma (RCC): no follow-up for a simple cyst, selective arterial embolisation (SAE) for a large AML, excision for a solid or complex cystic mass. Biopsy and SRM algorithm: *see Ch 137*.

2 · Renal cysts and the Bosniak classification

Why this matters: the Bosniak class decides between discharge, surveillance and surgery.

- A **simple cyst** (any size, homogeneous, **≤20 Hounsfield units (HU)**) needs no further imaging. Otherwise use Bosniak.

Class	Features (v2019)	Malignancy	Management (Canadian Urological Association, CUA 2023)
I	Thin (**≤2 mm**) smooth wall (may enhance); simple fluid; no septa or calcification	Benign	No follow-up
II	Thin wall plus (a) **≤3** thin enhancing septa, any calcification; or (b) homogeneous marked T2 hyperintensity; or (c) marked T1 hyperintensity	Highly likely benign	No follow-up
IIF	Minimally thickened (**3 mm**) enhancing wall or septum, or **≥4** thin septa; or heterogeneous T1 hyperintensity on fat-saturated magnetic resonance imaging (MRI)	Likely benign	Active surveillance up to **5 yr**
III	Enhancing thick (**≥4 mm**) or irregular (≤3 mm obtuse-margined protrusion) wall or septum	**~50%**	≤2 cm: surveillance preferred; 2–4 cm: surveillance or excision (shared decision); >4 cm: excision, partial (PN) over radical nephrectomy (RN) when feasible
IV	Enhancing nodule: ≥4 mm obtuse-margined, or any-size acute-margined protrusion	**~90%**	

KEY POINT

Ladder: wall ≤2 mm, then 3 mm, then ≥4 mm, then irregular, then nodule. Traps: any calcification fits II; a class I wall may enhance; large hyperdense cysts are no longer IIF; growth of I/II is not malignancy.

KEY POINT

Why surveil small III/IV? Class III has a low positive predictive value. Cystic RCC is indolent: mostly fluid, so size predicts poorly (proposed **pT1c**). The number needed to treat is therefore very high.

- European Association of Urology (EAU): treat Bosniak IV as a localised solid mass. Biopsy only a solid component, and only if it changes management. Ablation of cystic masses is unproven.

Diffuse cystic disease

- P2 ADPKD (Adult polycystic kidney disease); Acquired renal cystic disease — *genetics, extrarenal cysts, US criteria, tolvaptan, cyst complications; ACKD prevalence, ADPKD vs ACKD*

Not in the paper answer:

- Acquired cystic kidney disease (ACKD) cysts arise from the proximal tubule; **80%** have them by 10 yr of dialysis.
- The chapter puts ACKD RCC risk at **≥10×** (multifocal, bilateral, papillary commoner; a distinct WHO 2022 entity).
- In ADPKD, RCC is not commoner but is earlier, sarcomatoid and harder to diagnose.
- PKD1 relatives aged 15–29 need **≥3 cysts**; a normal scan excludes disease only from age 40.
- Confirm cyst infection with FDG-PET (fluorodeoxyglucose positron emission tomography). Bulk pain: aspiration, embolisation or fenestration.

3 · Oncocytoma

Why this matters: biopsy cannot reliably separate it from chromophobe RCC.

- The **commonest benign enhancing renal mass**: up to half of benign SRMs <4 cm.
- **Birt–Hogg–Dubé** (folliculin, chr 17) gives fibrofolliculomas, lung cysts with pneumothorax, oncocytoma and chromophobe RCC (chRCC). Oncocytosis (diffuse bilateral nodules) may harbour chRCC or hybrid tumours.
- A **stellate scar** is non-specific; **segmental enhancement inversion** favours oncocytoma.

	Oncocytoma	Chromophobe RCC
Cytokeratin 7 (CK7)	Rare, single-cell; KIT positive	Diffuse positive
Cytogenetics	Normal, −1, −Y, 11q13 rearrangement	Extensive abnormalities

KEY POINT

Why can a core not clear oncocytoma? Sampling error, and chRCC has oncocytic areas, so a minority of biopsy "oncocytomas" are RCC. EAU: no definitive core diagnosis; CK7 and cytogenetics help.

4 · Angiomyolipoma

Why this matters: AML can bleed catastrophically, so treatment follows bleeding risk.

- P3 P2 Management of AML; Renal aangiomyolipoma — *PEComa, TSC, mTOR, fat on CT, epithelioid variant, triggers, SAE, NSS, everolimus*

Not in the paper answer:

- Multiple AMLs are clonally independent; keratin negative.
- Giant AML arises from parenchyma; liposarcoma envelops the kidney.
- An observed AML ≥4 cm needs CT angiography to find an aneurysm.
- Haemorrhage in tuberous sclerosis complex (TSC): embolise, then give corticosteroids; avoid nephrectomy.
- EAU (sporadic): use a mammalian target of rapamycin (mTOR) inhibitor only if unfit for SAE or surgery.

ALGORITHM

1. **Spontaneous retroperitoneal haemorrhage** (Wunderlich†): resuscitate, then **SAE**. Nephrectomy only if SAE is unavailable or unsuitable.
2. **Symptomatic** (pain, haematuria, barriers to care): SAE, then nephron-sparing surgery (NSS) or RN; ablation in selected cases.
3. **Asymptomatic sporadic**: if > **6 cm**, aneurysm **≥5 mm**, growth > **0.25 cm/yr**, or childbearing age/pregnancy, then treat; otherwise surveil.
4. **TSC**: if > **3 cm** or growing, then everolimus first line; SAE or kidney-sparing resection second.

KEY POINT

Why abandon 4 cm? It is poorly specific, so it caused overtreatment; AML <4 cm is surveilled. "Barriers to care" counts because surveillance needs access to imaging and emergency care. In pregnancy, treat before conception.

5 · Other benign epithelial and mesenchymal tumours

Why this matters: these rare tumours mimic RCC.

- **Papillary adenoma** is unencapsulated and **≤15 mm**. It shares +7, +17, −Y with papillary RCC but cannot metastasise; acceptable in a donor kidney.
- **Metanephric adenoma** affects women in their 50s and may cause polycythaemia. Hypovascular; biopsy is unproven, so excise by NSS. **WT1, CD57 positive; CK7, AMACR negative**; lacks +7/+17/−Y.
- **Mixed epithelial and stromal tumour (MEST) and adult cystic nephroma** are one WHO 2022 family: middle-aged women, oestrogen/progesterone-receptor-positive stroma; manage by Bosniak. Paediatric cystic nephroma is distinct (**DICER1**).
- **Leiomyoma** arises from the renal capsule (desmin, caldesmon positive). It is often really a lipid-poor AML; **Melan-A** is positive only in AML.

Tumour	Key trait
Haemangioma	Sturge–Weber, Klippel–Trénaunay; usually excised
Haemangioblastoma	Von Hippel–Lindau (VHL); mimics RCC; cured by excision
Lymphangioma	Perinephric cysts; chyluria
Reninoma (juxtaglomerular cell tumour)	Young women; refractory hypertension, hypokalaemia, high renin; renin granules; excision cures
Solitary fibrous tumour	~10% aggressive: age ≥55, ≥15 cm, mitoses, necrosis

6 · Landmark trial

Trial	Design	Result	Take-home
EXIST-2	Phase III randomised trial, everolimus vs placebo, AML in TSC or lymphangioleiomyomatosis	≥50% volume reduction **42% vs 0%**; no bleeding needing embolisation	Approved; mTOR inhibition first line in TSC

137 Malignant Renal Tumours

1 · Overview, epidemiology & aetiology

Why this matters: most kidney cancers are now found by chance and many are indolent, so the main modern risk is overtreatment.

- Renal cell carcinoma (RCC) makes up 2–3% of adult cancers and is the most lethal common urological cancer. Male to female ratio is **2:1**, peak age 55–75 y, and **4–6% are familial**.
- Over 60% are incidental and ~70% are organ-confined. Incidence is rising faster than mortality, so avoid overtreatment.
- Risk factors: **tobacco** (best accepted; relative risk ~1.4, dose-dependent, falls after quitting); **obesity** ("obesity paradox": higher body mass index (BMI) but lower stage); **hypertension** (the disease, not its drugs); chronic kidney disease (CKD) and end-stage renal disease (ESRD); family history (2–4×).
- Clear cell and papillary RCC arise from the **proximal tubule**; chromophobe and collecting duct tumours from the distal nephron.
- In children and young adults, RCC is 1–6% of paediatric renal tumours (Wilms ~90%) and equals Wilms in the 2nd decade. The commonest type is **Xp11/TFE3 translocation RCC**. These patients do well with surgery even when node-positive, so do a formal lymph node dissection (LND) at radical nephrectomy (RN).

2 · Classification & pathology

Why this matters: subtype and grade drive prognosis, follow-up and which drugs work.

WHO 2022 classification (key groups)

Group	Entities
Renal cell tumours	Clear cell; multilocular cystic neoplasm of low malignant potential; papillary; chromophobe; other oncocytic; RCC not otherwise specified (NOS)
Molecularly defined (new)	TFE3/TFEB-rearranged; ELOC-mutated; fumarate hydratase (FH)-deficient; succinate dehydrogenase (SDH)-deficient; ALK-rearranged; SMARCB1-deficient medullary
Other	Collecting duct; mucinous tubular & spindle cell; tubulocystic; acquired cystic disease-associated; eosinophilic solid & cystic; clear cell papillary tumour
Non-RCC malignant	Urothelial/squamous; sarcomas; nephroblastic; neuroendocrine; lymphoma; **epithelioid angiomyolipoma**; metastases

KEY POINT

Trap: epithelioid angiomyolipoma (AML) is malignant; classic AML, papillary adenoma and oncocytoma are benign. The papillary type 1/2 split was **abandoned** because "type 2" was several diseases (MET-altered, MiT translocation, FH-deficient) with different prognoses. WHO therefore moved to molecular definitions.

Gross features

- RCC has a **pseudocapsule** of compressed parenchyma and is not infiltrative, except collecting duct and sarcomatoid tumours. Cystic degeneration occurs in 10–25% (better prognosis) and calcification in ~10%.

- About 20% are locally aggressive (perinephric fat, sinus, collecting system), but Gerota's fascia usually protects adjacent organs. Venous involvement occurs in ~10% (more than any other tumour) and can reach the right atrium.
- Sporadic RCC is bilateral in 2–4% and multicentric in 10–20%. Occult satellites are the main cause of ipsilateral recurrence after partial nephrectomy (PN).

Subtypes

Subtype (share)	Genetics	Hallmarks	Behaviour
Clear cell (70–80%)	3p loss/VHL (>90% 3p altered); +5q; PBRM1, SETD2, BAP1	Hypervascular; clear cytoplasm; carbonic anhydrase IX (CA-IX) positive; sarcomatoid 5–8%	Most venous invasion; worse when localised, better when metastatic
Papillary (10–15%)	**Trisomy 7, 17**, loss of Y; MET altered 81%	Hypovascular; CK7, AMACR; multicentric up to 40%	Good (5-y cancer-specific survival (CSS) 86–92%); common in acquired cystic disease
Chromophobe (3–5%)	Losses 1, 2, 6, 10, 13, 17; TP53, PTEN; FLCN in Birt–Hogg–Dubé (BHD)	"Plant cells", perinuclear halo, raisinoid nuclei; diffuse CK7	From intercalated cells; >90% cancer-free at 5 y unless sarcomatoid
Clear cell papillary tumour (<5%)	—	Linear papillae; CK7; **cup-like CA-IX**	Seen in ESRD and VHL; now benign
Collecting duct (Bellini)	Multiple losses	Central, infiltrative, desmoplastic; PAX8 positive; diagnosis of exclusion	Poor; urothelial-like, so may respond to cisplatin/gemcitabine
SMARCB1-deficient medullary	**Sickle cell trait**	Young Black patients; infiltrative	Dismal, median 8–13 mo; modest platinum response
FH-deficient	FH (1q42–43)	Macronucleolus with perinucleolar halo; FH loss, 2SC positive	Metastasises early
MiT translocation	Xp11.2 TFE3, TFEB	Clear cells with papillae; FISH confirms	40% of paediatric RCC; t(X;1) gives late nodal recurrence
SDH-deficient	SDH genes	Vacuolated eosinophilic cells; SDHB loss	Young; usually favourable; often germline
ALK-rearranged	ALK (2p23)	Test ALK before calling a tumour "unclassified"	Dramatic responses to ALK inhibitors in some
Unclassified (1–5%)	—	Shrinking catch-all	Poor

- **Indolent types:** multilocular cystic, tubulocystic, mucinous tubular & spindle cell, eosinophilic solid & cystic, ELOC, clear cell papillary. **Lethal types:** medullary, FH-deficient, collecting duct, unclassified, sarcomatoid.
- Chromophobe and oncocytoma are hard to separate, so WHO 2022 added "oncocytic renal neoplasm of low malignant potential". Papillary adenoma ≤15 mm is benign.
- **Sarcomatoid** change (1–5%) is dedifferentiation of any subtype; median survival is <1 y. Sarcomatoid or rhabdoid features mean automatic grade 4 and must be reported in every

RCC.

> **KEY POINT**
> Localised clear cell RCC does worse than papillary or chromophobe because it is intrinsically aggressive. Yet metastatic clear cell does better, because VHL-driven angiogenesis and immunogenicity make it respond to tyrosine kinase inhibitors (TKIs) and immune checkpoint inhibitors (ICIs).

WHO/ISUP (International Society of Urological Pathology) grading: replaced Fuhrman; validated for clear cell & papillary

Grade	Criterion
1	Nucleoli absent/inconspicuous, basophilic at ×400
2	Nucleoli conspicuous, eosinophilic at ×400; not prominent at ×100
3	Nucleoli conspicuous, eosinophilic at **×100**
4	Extreme pleomorphism, giant cells, **rhabdoid/sarcomatoid**

Nuclear grade is an independent prognostic factor.

3 · Molecular biology & hereditary syndromes

Why this matters: the VHL pathway explains why clear cell RCC is vascular and which drugs work; a hereditary syndrome changes surgery for the whole family.

- P2 P1 VHL gene and von Hippel–Lindau disease; Familial malignancies in urology — *VHL–HIF pathway, two-hit, VHL lesions, genotype–phenotype, VHL screening, 3-cm rule, belzutifan; VHL/HPRC/HLRCC/BHD/TSC genes and features*

Not in the paper answer:

- Multidrug-resistance pumps explain why chemotherapy response is only 5–7%; checkpoint blockade is now the mainstay (*see Ch 144*).
- Poor prognosis in all subtypes: **CDKN2A** (9p) loss, CpG hypermethylation and a Th2 signature.
- BAP1 mutation (up to 14% of sporadic clear cell) gives high grade and rapid growth, so these tumours are unsuitable for active surveillance (AS). Test germline BAP1 in early-onset, multifocal, VHL-negative clear cell RCC.
- Syndromes beyond the paper answers are in the table below. Hereditary papillary RCC is indolent, so avoid CKD from repeated surgery.

> **ALGORITHM**
> 1. Normoxia: the VHL protein joins elongin B/C, cullin-2 and RBX1 to form an E3 ligase, which degrades hypoxia-inducible factor (HIF)-1α/2α.
> 2. If VHL is lost (or in hypoxia), **HIF-2α** accumulates and pairs with ARNT. This raises VEGF (vascular endothelial growth factor), PDGF, TGF-α, GLUT1, erythropoietin (EPO) and CA-IX → hypervascular clear cell RCC.

Drug targets: VEGF ligand (bevacizumab); VEGF-receptor TKIs (sunitinib, pazopanib, axitinib, cabozantinib); mammalian target of rapamycin (mTOR) (temsirolimus, everolimus); HIF-2α (**belzutifan**, approved by the US Food and Drug Administration (FDA) in 2021 for VHL, ~50% response); CTLA-4 (ipilimumab); PD-1 (nivolumab, pembrolizumab); PD-L1 (atezolizumab, avelumab).

Syndrome (beyond topics)	Gene (chr)	Renal tumour	Other / management
Hereditary leiomyomatosis and RCC (HLRCC)	FH (1q42–43)	Aggressive, often **solitary**	Uterine leiomyosarcoma; **prompt excision at any size, wide margins**; PN only if a wide margin is feasible; penetrance 15–20%; fluorodeoxyglucose PET (FDG-PET) finds small metastases
SDH-RCC	SDHA/B/C/D, SDHAF2	Onset ≤40 y	Paraganglioma/phaeochromocytoma, gastrointestinal stromal tumour, thyroid; excision, PN preferred
Cowden	PTEN (10q23)	Mostly papillary (lifetime 34%)	Breast (50%), thyroid; KILLIN if PTEN-negative; CT/MRI, 3-cm rule
Tuberous sclerosis	TSC1/TSC2	AMLs; clear cell 1–3%	AML: surveil <3 cm, everolimus 3–5 cm, embolise or excise ≥5 cm
BAP1	BAP1 (3p21)	High-grade clear cell	Uveal/cutaneous melanoma, mesothelioma; nephron-sparing excision
MiTF-associated	MiTF p.E318K	RCC >5× (activates HIF-1α)	Melanoma; management undefined

4 · Clinical presentation

Why this matters: symptoms usually mean advanced disease, and paraneoplastic signs can be the first clue.

- The **classic triad** (flank pain, gross haematuria, mass) is rare and means advanced disease (the "too late triad"). Other clues: weight loss, bone pain, cervical nodes, a **non-reducing or right-sided varicocele**, and leg oedema from inferior vena cava (IVC) obstruction.
- Over 50% of spontaneous perirenal haematomas are due to a tumour, so repeat the CT once the blood resolves.

Paraneoplastic syndromes (10–20%)

Syndrome	Frequency / mechanism	Key point
Raised erythrocyte sedimentation rate (ESR)	Commonest	Marker of advanced disease
Hypercalcaemia	Up to 13%; parathyroid hormone-related protein (PTHrP) or osteolytic metastases	**The only one managed medically**: hydration, then furosemide; zoledronic acid 4 mg IV 4-weekly (withhold in renal insufficiency); denosumab if refractory; steroids, calcitonin. Definitive: nephrectomy with or without metastasectomy; the osteolytic type is harder (bisphosphonates, radiotherapy)
Hypertension	Renin, artery compression, arteriovenous fistula	—
Polycythaemia	Erythropoietin	—

Syndrome	Frequency / mechanism	Key point
Stauffer syndrome	3–20%; interleukin-6	Non-metastatic liver dysfunction (alkaline phosphatase (ALP) raised almost always); exclude liver metastases; normalises after nephrectomy in 60–70%; persistence means viable tumour and poor prognosis
Others	Rare	Cushing, hyperglycaemia, galactorrhoea, neuromyopathy, clotting disorders, cerebellar ataxia

- Rule: paraneoplastic syndromes resolve only with excision or systemic therapy (except medically managed hypercalcaemia). Persistence after nephrectomy means residual disease.

5 · Diagnosis

Why this matters: imaging sorts a mass into cyst, fat (AML) or enhancing solid tumour, and biopsy is used only when it changes the plan.

- P4 Radiology for solid renal masses — *CT phases and HU thresholds, MRI, CEUS, sestamibi, girentuximab, FDG-PET, biopsy yield, venous thrombus CT sensitivity, R.E.N.A.L./PADUA*

Not in the paper answer:

- The strongest predictors of malignancy are **male sex (~3×)** and **size**. Benign: 20–30% of masses ≤4 cm and ~40% of ≤2 cm. At ≥7 cm few are benign and half are high grade.
- Nomograms barely beat a coin toss (concordance 0.55–0.65). A cT1 mass is ~20% benign, ~60% indolent and ~20% potentially aggressive.
- If iodinated contrast is given with estimated GFR (eGFR) <45, give periprocedural hydration. Macrocyclic gadolinium is safe even on dialysis.
- An infiltrative pattern suggests lymphoma, urothelial carcinoma, sarcomatoid or collecting duct RCC, medullary carcinoma, or xanthogranulomatous pyelonephritis.
- Renal mass biopsy (RMB): **multiple cores over fine-needle aspiration**. Haematoma ~5%, transfusion <0.5%, specificity 94%. A non-malignant result does not prove the mass is benign.
- **Multiphase CT** is the single most important test: enhancement **>15–20 Hounsfield units (HU)** suggests RCC, and fat **<−20 HU** means AML. MRI is best for IVC thrombus and organ invasion.

ALGORITHM

1. If simple cyst, then observe.
2. If complex or solid: eGFR >45 → CT or MRI with/without contrast; eGFR 30–45 → hydrate, then the same; eGFR <30 → non-contrast CT or gadolinium MRI; contrast allergy → ultrasound (USG), contrast USG or MRI.
3. Then: fat → AML; complex cyst → Bosniak; solid enhancing → management options, with or without RMB.

- **Bosniak** (*see Ch 136*): malignancy is 3–10% in IIF, ~50% in III and 75–90% in IV. The key discriminator is unequivocal **measurable enhancement**. Avoid ablation and biopsy of cystic masses (poor yield, spillage); AS is acceptable for III/IV.
- Do RMB when the result may change management, or when lymphoma, metastasis or infection is suspected. Do not biopsy patients who will be managed the same way regardless. Biopsy before ablation.
- **Screening:** none for the population or for autosomal dominant polycystic kidney disease. In ESRD (acquired cystic disease in 80%; RCC in 1–2% of these), screen with USG or CT from

the 3rd year of dialysis if fit, and continue after transplant.

- Offer genetic counselling for RCC at **≤46 y**, multifocal or bilateral tumours, or a suggestive history or pathology.

6 · Staging & prognosis

Why this matters: pathological stage is the strongest predictor of outcome and sets the follow-up schedule.

TNM (AJCC 8th)	Definition	5-y survival
T1a / T1b	≤4 / >4–7 cm, confined	95–100 / 85–95%
T2a / T2b	>7–10 / >10 cm	65–85 / 55–75%
T3a	Renal vein/segmental branches, **pelvicalyceal system**, perirenal/sinus fat, within Gerota's	45–75%
T3b	IVC below diaphragm	40–55%
T3c	IVC above diaphragm or **caval wall invasion**	20–45%
T4	Beyond Gerota's, incl. **contiguous ipsilateral adrenal**	0–35%
N1 / M1	Regional node / distant (non-contiguous adrenal = M1)	0–30%

Stage groups: I is T1N0; II is T2N0; III is T1–2N1 or T3 any N; IV is T4 or M1. History: the 6th edition split T1 at 4 cm. The 7th split T2 at 10 cm, moved contiguous adrenal to T4 and renal vein thrombus from T3b to T3a, and merged N1/N2. The 8th moved pelvicalyceal invasion to T3a.

> **KEY POINT**
> TNM limitation: stage is set by the single worst feature, so co-existing adverse features (e.g. adrenal invasion plus a supradiaphragmatic thrombus) are lost. Record all of them and use integrated nomograms.

- **Work-up:** abdominal CT and chest X-ray (chest CT if symptoms or high risk). Bone scan only with raised ALP, bone pain or poor performance status (PS); no PET. Small nodes are often reactive and should not preclude surgery. Treat an abnormal adrenal as involved.
- **Pathological stage** is the most important factor, then grade, size, subtype, necrosis and sarcomatoid change. Node-positive 5-y survival is 5–30%. With a thrombus but otherwise confined disease, 5-y survival is 45–69%.
- Venous wall invasion is as important as thrombus level. Sinus fat invasion is possibly riskier than perinephric, and collecting-system invasion is adverse.
- Adverse clinical features: symptoms, weight loss >10%, poor PS, anaemia, hypercalcaemia, raised ESR, lactate dehydrogenase (LDH) or ALP, and thrombocytosis.
- A post hoc ASSURE analysis showed algorithms barely beat stage and grade, which led to the American Urological Association (AUA) risk groups. † **UISS** uses TNM, grade and ECOG PS; **SSIGN** (Mayo) uses Stage, Size, Grade and Necrosis; **Leibovich** uses pT, pN, size, grade and necrosis to predict recurrence in clear cell RCC.

7 · Management of localised RCC

Why this matters: the choice between surveillance, ablation, partial and radical nephrectomy is a trade-off between cancer control and kidney function.

- P1 P2 Embryology and anatomy of kidney; nephron-sparing surgery in small renal mass and renal malignancy; Nephron-sparing surgery / partial nephrectomy in RCC — *imperative/relative/elective PN, AUA RN rule (GFR >45), surgical vs medical CKD, PN steps, enucleation, solitary kidney, AS/TA criteria, 3-cm rule, EORTC 30904, DISSRM*

Not in the paper answer:

- AUA 2021: refer to nephrology if GFR <45, confirmed proteinuria, diabetic CKD, or expected post-op GFR <30. Do LND for clinically positive nodes and adrenalectomy only if suspicious. Examine adjacent parenchyma for medical renal disease.
- About 25% of "ideal" PN candidates already have CKD. CKD follows RN in 65% vs PN in 20%. Surgical CKD progresses ~0.7%/y vs medical CKD ~4.7%/y.
- The PN survival advantage in retrospective series is selection bias. Compensatory hypertrophy is 20–30%. **Predicted GFR after RN ≈ global GFR × contralateral split function × 1.25**.
- AS: small masses grow 0.1–0.3 cm/y and metastasise in <2%. Image every 3–6 mo on AS, or 6–12 mo on expectant (often palliative) management.
- **Triggers to intervene**: >3 cm, growth >5 mm/y, stage progression, adverse biopsy, symptoms. A fast-growing subgroup may lose the PN or thermal ablation (TA) window.
- PN function depends first on parenchymal quality (fixed), then on the **amount of vascularised parenchyma preserved (primary, modifiable)**, then on ischaemia (secondary).
- After PN, leak and bleeding are commoner than after RN; laparoscopic/robotic ≈ open. Its 1–2% local recurrence is balanced by 1–2% contralateral RCC. Steps: *see Ch 142*.
- Solitary kidney: ESRD ~4.5%; at least 20–30% of one kidney is needed. After >50% loss, give an angiotensin-converting enzyme (ACE) inhibitor and a low-protein diet.
- If bilateral tumours make PN impossible: RN with dialysis (transplant after a cancer-free interval), or neoadjuvant TKI (stop it several half-lives around surgery).
- Enucleation is controversial in sporadic RCC because 40–50% infiltrate the pseudocapsule (rarely low grade, chromophobe or oncocytoma), so RMB helps. Report technique by the surface–intermediate–base (SIB) protocol.

Radical nephrectomy

- Robson's principles were early artery then vein ligation, removal outside Gerota's fascia, adrenalectomy and extended LND.
- Today perifascial dissection is still essential (~25% of cT1b/T2 invade fat). There is **no routine adrenalectomy** (unless enlarged on imaging or extensive tumour) and **no routine extended LND**.
- Open: subcostal (preferred) or thoracoabdominal (huge upper-pole tumour). Laparoscopic RN is standard without gross invasion, major venous involvement or bulky nodes, but it has driven RN overuse (*see Ch 141*).

Thermal ablation (*see Ch 143*)

- TA works best <2.5–3 cm; use hydrodissection near bowel or hilum. There is no pathological staging.
- Cryoablation is lethal at ≤−20 °C, uses a double freeze–thaw, and the iceball extends ~1 cm beyond the tumour. Neither cryoablation nor radiofrequency is superior.

- Recurrence (table below) is mostly salvaged by repeat TA, with no difference in metastasis or CSS. Nodular enhancement means recurrence.

Familial RCC: 3-cm rule (National Cancer Institute)

- VHL tumours <3 cm had no metastases at ~5 y; ~25% metastasise when larger. **HLRCC is the exception**.
- Recurrence is near-universal (cyst linings), so surveillance is lifelong. **Exclude phaeochromocytoma** and central nervous system haemangioblastoma before surgery. Transplant after removing all renal tissue is safe.

8 · Locally advanced RCC

Why this matters: even with a caval thrombus, complete surgery can still cure.

IVC tumour thrombus (venous involvement 4–10%; cure 45–70%)

Level	Extent	T	Vascular control
Renal vein	Vein/branches	T3a	Standard RN
I	At renal vein ostium	T3b	Satinsky clamp/stapler; keep ≥50% lumen; patch if needed
II	Below hepatic veins	T3b	Clamp infrarenal IVC, then contralateral renal vein, then suprarenal IVC; occlude lumbars
III	Intrahepatic, below diaphragm	T3b	Mobilise liver and milk thrombus below hepatic veins; or suprahepatic clamp with Pringle; venovenous bypass often
IV	Above diaphragm	T3c	Cardiopulmonary bypass with or without hypothermic arrest; avoid opening the heart if thrombus can be brought below the atrium
Any, wall invasion	Caval wall	T3c	Resect to negative margins; graft/reconstruct

- Image close to surgery, because the thrombus grows. **No routine renal artery embolisation** (more deaths). Do coronary angiography before bypass (levels III–IV) and use intraoperative transoesophageal echocardiography.
- An occluded IVC may need no reconstruction because collaterals have formed. Bland distal thrombus is removed, ligated or filtered.
- Level and wall invasion predict survival; level IV without metastasis is curable. Mortality is 5–10% for supradiaphragmatic thrombus.

Locally invasive (T4)

- Only ~40% of suspected invasion is confirmed. En bloc resection is the only cure, but prognosis is poor. Debulking is rarely worthwhile (1-y survival 10–20%).

Lymph node dissection

- Only 30–43% of enlarged nodes are malignant. Risk factors are ≥10 cm, cT3/T4, grade 3–4, sarcomatoid change and necrosis; with ≥2 factors, >40% are node-positive.
- LND gave no survival benefit (EORTC 30881), so it is done for suspicious nodes, mainly for staging. Resect bulky nodes if feasible.

9 · Recurrence, adjuvant therapy & follow-up

Why this matters: a third of recurrences come late, and only one adjuvant drug has proven benefit.

Local recurrence	After RN	After PN	After TA
Rate	2–4%	1.4–10%	Cryo 3–10%, radiofrequency ablation 5–20%
Site	Fossa, adrenal, vein stump, nodes	Remnant, usually away from bed	Ablation zone
Management	Only 20–40% isolated, so do a full work-up; en bloc resection gives 30–50% long-term cancer-free	Repeat PN, completion nephrectomy, TA, AS	Repeat TA, AS, salvage surgery

- A positive PN margin is usually surveilled; completion nephrectomy only for gross involvement.
- Rationale for adjuvant therapy: 20–35% metastasise after "curative" surgery. Cytokines, vaccines, girentuximab and most TKIs failed. **Adjuvant pembrolizumab** is FDA-approved for high risk.
- Neoadjuvant TKI gives 10–30% shrinkage, with ~20% serious adverse events.

Follow-up (AUA risk groups)

Risk	Definition	Chest imaging
Low	pT1 grade 1–2	X-ray
Intermediate	pT1 grade 3–4 or pT2	X-ray
High	pT3	CT
Very high	pT4, pN1, sarcomatoid/rhabdoid, gross positive margin	CT

- Imaging starts at 3–6 mo and becomes more intensive with risk; very high risk is followed for up to 10 y. A microscopic positive margin moves the patient up one level. **30% of recurrences occur after 5 y**.
- Benign pathology needs no imaging. After TA, do contrast imaging within 6 mo, then follow the intermediate schedule.

10 · Other renal malignancies

Why this matters: some non-RCC tumours need surgery while others (lymphoma, metastasis) need biopsy and drugs, not nephrectomy.

Tumour	Clues	Management
Sarcoma (1–2%)	**Leiomyosarcoma commonest** (50–60%); capsule/sinus origin, huge, no nodes, hypovascular	Wide en bloc excision, negative margins; the first resection is the best chance; margin and grade give prognosis
Lymphoma	Non-Hodgkin; multiple masses (45%); massive adenopathy, splenomegaly; prior lymphoma means recurrence until proven otherwise	**Biopsy, avoid nephrectomy**; R-CHOP with or without radiotherapy
Metastases	Commonest renal malignancy (12% at autopsy); lung primary commonest; multifocal, 5–30 HU	Biopsy; systemic therapy; nephrectomy only for refractory bleeding

Tumour	Clues	Management
Carcinoid	**Horseshoe kidney (82×)**	Excision; good prognosis
Small cell	Usually advanced	Surgery and platinum
Primitive neuroectodermal tumour	Ewing family; CD99; t(11;22)	Multimodal; poor
Adult Wilms	3% of Wilms; triphasic	Paediatric protocols; worse than in children

- In sarcoma, any RCC feature excludes it (the differential is sarcomatoid RCC). Haemangiopericytoma is vascular, so embolise before surgery. Leiomyosarcoma is commoner than liposarcoma, the reverse of the retroperitoneum. Pre-op radiotherapy may reduce positive margins.
- Lymphoma is commoner with immunosuppression. Renal failure comes more from hypercalcaemia or urate than from tumour replacement.

> **KEY POINT**
> Non-RCC malignancies are almost all **hypovascular**. The key decision is surgical (sarcoma, carcinoid) vs medical (lymphoma, metastasis: biopsy, don't resect). RCC nodes are interaortocaval (right) or para-aortic (left); nodes elsewhere suggest lymphoma. A solitary enhancing mass years after an organ-confined non-renal cancer is more likely a primary RCC.

11 · Landmark trials

Trial	Design	Result	Take-home
EORTC 30904	Randomised trial (RCT), RN vs elective PN, ≤5 cm, normal contralateral kidney	PN better function; 10-y overall survival (OS) better with RN (81 vs 76%)	Flawed; PN's functional gain may be smaller than believed
EORTC 30881	RCT, RN with or without LND	No survival benefit	Underpowered, low-risk (4% node-positive); LND for staging
S-TRAC	Sunitinib vs placebo, high-risk clear cell	Disease-free survival (DFS) 6.8 vs 5.6 y; no OS gain	Only positive adjuvant TKI trial
KEYNOTE-564	Pembrolizumab vs placebo, high-risk	DFS hazard ratio 0.63	FDA 2021; only positive adjuvant ICI trial
DISSRM	Registry, AS vs intervention, small masses	1 death/metastasis on AS; 24% delayed intervention	AS safe for small masses

138 Urothelial Tumours of the Upper Urinary Tract and Ureter

1 · Epidemiology & risk factors

Why this matters: UTUC is often invasive at diagnosis and its causes act as a field effect.

- P3 P2 Upper Tract Urothelial Carcinoma: Diagnosis, Staging, Minimally Invasive, Endoscopic and Radical Management; Etiopathogenesis and management of Upper tract TCC — *epidemiology, tobacco/aristolochic acid/Lynch/occupation/phenacetin/arsenic, seeding vs field effect, CTU, cytology, URS biopsy, TNM, EAU risk groups, management*

Not in the paper answer:

- Bilateral ≤5%; multifocal **10–20%**; concomitant carcinoma in situ (CIS) 11–36%; 7–9% metastatic at diagnosis; contralateral upper tract recurrence 2–6%.
- Stopping smoking for >10 yr cuts the risk by 60–70%; smoking risk is higher for ureter than pelvis.
- Aristolochic acid (Belgium 1992–93 slimming herbs): no sex bias, onset ≈10 yr earlier, TP53 codon 139. Arsenic: female 2:1, ureter 2:1, causal link unproven. Phenacetin: latency ≈22 yr; pelvis more than ureter.
- Lynch syndrome (LS): mean age 55, relatively more women, upper tract more than bladder.
- **EAU: screen for LS at the first interview** if age <60, a personal LS-spectrum cancer, a first-degree relative (FDR) <50 with LS cancer, or two FDRs with LS cancer. Then germline sequencing, colorectal/endometrial surveillance, family counselling.
- Upper tract urothelial carcinoma (UTUC) is **5–10%** of urothelial cancers; in the Balkans it is **40%** of renal neoplasms. Concurrent bladder cancer occurs in **17%**; ~60% are invasive at diagnosis.
- After UTUC, the bladder recurs in **22–47%**.

> KEY POINT
> **Why UTUC presents invasive:** the muscle layer is thin (mesodermal), so tumours reach muscle, parenchyma or periureteric fat earlier than bladder tumours.

- **Smoking** carries a relative risk (RR) of **2.5–7**. Occupational exposure averages ≈7 yr. Chronic infection or stones carry RR 1.5–2 (squamous cell carcinoma, SCC).
- Iatrogenic causes: cyclophosphamide (acrolein), anthranoid laxatives for >1 yr, pelvic radiotherapy.

> KEY POINT
> Aristolochic acid, arsenic, phenacetin and Lynch all behave as a **field disease** (multiple, bilateral recurrences). This makes kidney-sparing treatment more attractive, but Balkan endemic nephropathy (BEN) patients often have renal insufficiency. Analgesic abuse history is present in 22% of pelvic and 11% of ureteric tumours.

- **Lynch**: UTUC risk is ≈**14×**, and ≈**75×** in MSH2 carriers. If mismatch repair (MMR) proteins are lost on immunohistochemistry (IHC), do germline testing.

2 · Pathology & molecular biology

Why this matters: some histologies are aggressive.

- Over 90% are urothelial carcinoma (UC). Variant histology (≈**25%**: squamous, glandular, sarcomatoid, micropapillary, neuroendocrine, lymphoepithelial) is aggressive;

micropapillary is independently adverse. CIS is easily missed.

- Ureteric inverted papilloma harbours malignancy in ≈**18%**, so follow it for ≥2 yr.
- SCC is 0.7–7%, 6× commoner in the pelvis, and invasive at presentation. Adenocarcinoma is <1%, linked to stones and obstruction, with poor prognosis. Sessile tumours invade in >80%.
- UTUC and bladder cancer are genetically distinct but clonally related. The **RAS-mutated** subtype does worst. Microsatellite instability (MSI) and hypermethylation distinguish UTUC from bladder cancer; MSI is prognostic and flags LS. Uroplakin III and p27 expression are favourable.

3 · Diagnosis

Why this matters: biopsy gives grade but not stage.

- Haematuria is commonest; then flank pain; mass ≈10%; incidental ≈15%. **Always do cystoscopy**. Positive cytology with normal cystoscopy (including urethra) means the upper tracts must be evaluated.
- **CT urography (CTU)** is first line (EAU prefers it to MR urography (MRU)): sensitivity ≈**92%**, specificity ≈95%. A filling defect is seen in 50–75%; flat lesions are missed. UC is ≈46 HU vs ≈100 HU for a uric acid stone. Hydronephrosis means an invasive ureteric tumour in up to 80%.
- MRU: avoid gadolinium if creatinine clearance <30 mL/min. Retrograde pyelography ≈75%. Do a split-function scan before radical surgery.
- Cytology ≈65% overall (G1 20%, G2 45%, G3 75%). Selective washing beats voided urine; brush plus cytology ≈90%. **Collect before contrast**. FISH is best for low-grade surveillance, not routine.
- Flexible ureteroscopy (URS) with biopsy raises accuracy from 75–85% to ≈90%. Biopsy grade is ≈90% accurate; visual grading only ≈70%.

> **KEY POINT**
> **Why URS biopsy cannot stage:** samples are small and shallow, so a Ta biopsy is often T3 at resection. Stage is **inferred** from grade, appearance, hydronephrosis and cytology, which is why kidney-sparing patients need intensive follow-up.

- **Diagnostic URS indications:** uncertainty, planned kidney-sparing treatment, solitary kidney. **Pitfall:** URS before radical nephroureterectomy (RNU) increases intravesical recurrence (especially pelvic tumours and biopsy) without changing survival. Avoid it when CTU and cytology already justify RNU.

4 · Staging & grading

Why this matters: pelvic T3 vs T4 is a classic trap.

TNM 8th	Definition
Ta / Tis	Non-invasive papillary / CIS
T1	Subepithelial connective tissue
T2	Muscularis
T3	Pelvis: peripelvic fat or **renal parenchyma**; ureter: periureteric fat. pT3a = microscopic parenchymal; pT3b = macroscopic or peripelvic fat (worse)

TNM 8th	Definition
T4	Adjacent organs, or through kidney into perinephric fat
N1 / N2	Single node ≤2 cm / single >2 cm or multiple

- AJCC stages: I = T1; II = T2; III = T3; **IV = T4, any N+ or M1**.
- Pelvic and upper ureteric tumours drain to hilar and retroperitoneal nodes; lower ureteric tumours to pelvic nodes. WHO 2004/2016 grades: PUNLMP (managed as low grade), low and high grade.

5 · Prognosis & risk stratification

Why this matters: the risk group decides between kidney-sparing treatment and RNU.

- **Stage matters most**, then grade. SEER 5-yr cancer-specific survival after RNU: ≤T1 86%, pT2 77%, pT3 63%, pT4/N+ 39%.
- **Ureteric and multifocal tumours do worse** than pelvic ones at the same stage, because there is no parenchymal barrier and periureteric tissue has a rich lymphovascular plexus.
- Pre-op factors: age (never alone a reason to deny RNU), smoking, ASA/ECOG status, obesity, low albumin, high NLR/CRP. Do **RNU within 12 wk** for invasive disease. Size **2 cm** is the best cut-off for ≥pT2.
- Post-op factors: nodes (up to 40%), extranodal extension, node density ≥30%, lymphovascular invasion (LVI ≈20%; never seen on biopsy, so report it on every specimen), margins, necrosis >10%, sessile growth, CIS, variant histology.

Low risk — ALL	High risk — ANY ONE (EAU 2020)
Unifocal	Multifocal
<2 cm	>2 cm
Low-grade cytology	High-grade cytology
Low-grade URS biopsy	High-grade URS biopsy
No invasive aspect on CTU	Hydronephrosis
—	Previous radical cystectomy; variant histology

KEY POINT
Every pre-op factor is a stand-in for invasion, so one adverse feature means high risk. **Trap:** a small, low-grade, unifocal tumour **with hydronephrosis is high risk**.

6 · Recurrence patterns

Why this matters: recurrence runs both ways.

- Bladder recurrence after UTUC is 15–75% at 5 yr, usually within 2 yr.
- Upper tract tumours after bladder cancer: 2–4%, late (median >3 yr), and 0.75–6.4% after cystectomy. About **30%** of bacillus Calmette–Guérin (BCG)-treated bladder CIS (or cystectomy for BCG-refractory disease) develop UTUC. Risk factors: distal ureter/orifice tumour, multifocal or recurrent non-muscle-invasive bladder cancer, CIS, reflux.

KEY POINT

Asymmetric recurrence: antegrade seeding makes bladder recurrences common, early and monoclonal; the field effect gives multifocal tumours of different clones. Both operate.

- Intravesical recurrence after RNU: patient factors (male, prior bladder cancer, smoking, chronic kidney disease), tumour (positive cytology, ureteric, multifocal, invasive, necrosis) and treatment (laparoscopic, extravesical cuff, positive margins, diagnostic URS).
- **Pan-urothelial disease** means bladder plus ≥2 extravesical sites (women: both upper tracts; men: upper tract(s) and/or prostatic urethra). There is no evidence-based algorithm: cystectomy for high-grade bladder disease, conservative care for non-infiltrating UTUC, total urinary tract removal in the young. Cystectomy then RNU may lose cisplatin eligibility.

7 · Management overview & follow-up

Why this matters: risk group picks the operation.

ALGORITHM

1. Work-up: cystoscopy and CTU, with or without URS biopsy and cytology; assess contralateral function; screen for LS; then assign the risk group.
2. If **low risk**, then kidney-sparing treatment (endoscopic or segmental) with intensive surveillance.
3. If **high risk**, then RNU with bladder cuff, with or without LND, within 12 wk.
4. If **cN+**, then systemic chemotherapy first, and surgery only if there is a response.
5. After RNU, give a single intravesical chemotherapy dose; all patients have cystoscopic follow-up.

- Pre-op nomograms (≥pT2 or non-organ-confined) select patients for neoadjuvant chemotherapy; pathological downstaging improves overall survival (OS). Give chemotherapy before RNU because nephrectomy often removes cisplatin eligibility.
- Lymph node dissection (LND) gives the best staging; its curative role is debated. Technique and POUT: *see Ch 139*.
- Follow-up: cystoscopy for all. After kidney-sparing treatment, add URS, in-situ cytology and CTU. After high-risk bladder cancer (BCG-treated CIS, BCG-refractory), do yearly upper-tract CT (replacing cytology and excretory urography). Risk is constant, so surveillance is prolonged.

139 Surgical Management of Upper Urinary Tract Urothelial Tumours

Epidemiology, risk factors, pathology, staging and imaging: *Ch 138*

1 · Principles & patient selection

Tumour risk and renal reserve together decide between removing and saving the kidney.

- P3 Upper Tract Urothelial Carcinoma (including TCC of Ureter): Diagnosis, Staging, Minimally Invasive, Endoscopic and Radical Management — *AUA/SUO favourable/unfavourable groups and ablation role, elective vs imperative KSM, URS and percutaneous steps, lasers, segmental options, RNU principles, abandoned cuff techniques, topical BCG/UGN-101, follow-up*

Not in the paper answer:

- ~3× reinterventions after endoscopy; staged biopsy; expectant management; CIS work-up (§4); recurrence 33–55% and reconstruction numbers (§3); no routine adrenalectomy, lateral lower-ureteric supply, lasso abandoned, intraoperative instillation (§6–7); BCG granulomas (§5).
- **Standard** treatment of upper tract urothelial carcinoma (UTUC) is radical nephroureterectomy (RNU) with bladder cuff, if the other kidney is normal and the patient fit.
- UTUC is **unforgiving of spillage**, so every approach keeps the collecting system closed.
- Work-up: computed tomography (CT) or magnetic resonance (MR) urogram, then cystoscopy, then ureteroscopy (URS) with biopsy and selective cytology. URS raises accuracy from ≈75% to 85–90%.
- Without retrograde access (diversion), use antegrade endoscopy; if endoscopy is unsafe, use image-guided needle biopsy.
- Keep diagnostic endoscopy brief and low-pressure, because it can cause seeding and pyelovenous or pyelolymphatic migration.
- Low-grade (LG) tumours are mostly Ta/T1; high-grade (HG) tumours are mostly invasive.

AUA/SUO* risk group	Features	Ablation	Systemic therapy
Low – favourable	LG, negative cytology, unifocal, papillary, no obstruction/invasion	**Preferred**	No
Low – unfavourable	LG with obstruction, multifocality or prior lower-tract tumour	May be offered	No
High – favourable	HG, unifocal, papillary, no invasion/obstruction	Rare, select cases	Neoadjuvant/adjuvant
High – unfavourable	HG cytology, invasion on imaging, obstruction, suspicious nodes, multifocal, sessile/flat	Palliation only	Neoadjuvant/adjuvant

*American Urological Association / Society of Urologic Oncology.

KEY POINT
Low grade does not mean automatic ablation: obstruction or multifocality makes an LG tumour unfavourable, where RNU keeps a role.

KEY POINT
Two axes: tumour risk and renal reserve. High risk with adequate reserve means extirpation; low risk means preservation. Imperative indications are where they conflict.

ALGORITHM

1. Imaging suspicion: CT or MR urogram.
2. Endoscopy with biopsy and cytology; assign risk group; shared decision.
3. If high risk with adequate glomerular filtration rate (GFR), then RNU with or without neoadjuvant chemotherapy.
4. If low risk or poor GFR, then kidney-sparing management (KSM).

- Imperative KSM must not significantly compromise cancer control; dialysis risk is weighed against oncological risk.
- Lower ureteric tumours, especially unfavourable ones, need distal ureterectomy, not endoscopy.
- **Expectant management** (AUA/SUO option) is only for major comorbidity, competing mortality or high risk of end-stage renal disease (ESRD). Untreated UTUC progresses, with markedly worse survival.

2 · Endoscopic (kidney-sparing) management

Endoscopy saves the kidney but recurrence is expected, so selection matters.

- **Grade** is the key selector and the strongest predictor of recurrence; HG results are poor. AUA/SUO: tumours **<1.5 cm**; caution at ≥1.5 cm.
- For a bulky or suspected HG tumour, use a **staged approach**: biopsy and partial ablation only, avoiding perforation before a possible RNU; complete it once pathology confirms suitability.
- Versus RNU, survival is similar but there are **~3× more reinterventions**, with lifelong surveillance. Papillary and fornix tumours are hard to treat safely.

	Retrograde ureteroscopic	Antegrade percutaneous
System	**Closed**	**Open** – tract seeding ≈0.3%, all HG
Best for	Low-volume ureteric, small renal	Large pelvic/upper ureteric, lower-pole calyx, diversion
Staging	Not possible	Possible – base sent separately
Morbidity	Low	Higher (≈25%), admission
Recurrence	Higher bladder and upper-tract	Lower
Second look / topical therapy	Repeat URS; catheter or stent reflux	Via retained nephrostomy

- **URS:** saline barbotage improves cytology. Pre-stent a tight ureter (passive dilation) rather than dilating forcefully. Access sheath: no recommendation.
- Fix the minute specimens at once; adding selective cytology to biopsy improves grading.
- Use forceps or basket excision with base ablation. Ureteroscopic resection never goes beyond the lamina propria. The usual method is a **3-Fr biopsy, then laser**.
- A tumour protruding from the orifice needs ablation or resection of the whole intramural ureter.
- Bugbee depth is unpredictable. **Never fulgurate circumferentially**, because a stricture follows.
- Lasers: holmium:YAG (Ho:YAG) <0.5 mm, ideal for the ureter; neodymium:YAG (Nd:YAG) 5–6 mm, with transmural risk; thulium <0.2 mm, with strong coagulation.

- **Percutaneous:** puncture the calyx distal to a peripheral tumour, never through it; use an upper or middle calyx for the pelvis. Dilate to 30 Fr; the tract may mature for 1–2 weeks.
- Resect with cold-cup forceps (thin stalk) or a **bipolar** resectoscope, because monopolar risks hypo-osmotic absorption.
- A **nephrostomy is always left** for second look and topical therapy. HG or invasive histology means RNU.
- In a diversion, know the anastomosis type. Try flexible endoscopy first; otherwise pass an antegrade wire into the conduit and do retrograde URS over it.

> **KEY POINT**
> Does URS spread tumour? Prior ureteroscopy increases bladder recurrence (seeding), but pre-RNU URS has not increased metastasis. A single post-op intravesical instillation mitigates this.

3 · Segmental ureterectomy

Excising only the diseased ureter can match RNU.

- It saves nephrons **and** gives a staging specimen. In selected patients it equals RNU, because stage and grade set prognosis.
- **URS of the entire tract above is mandatory**. For HG, add ipsilateral pelvic lymph node dissection (LND).
- Ipsilateral recurrence is **33–55%**, mostly distal to the original site, and lifelong.
- Segmental resection with uretero-ureterostomy is rare: for an LG proximal or mid tumour that cannot be ablated, or HG when the unit must be saved. Use 1–2 cm margins and frozen section; spatulate, stent, keep it tension-free.
- Distal ureterectomy with reimplant suits LG low-stage tumours, or selected HG distal ones when the kidney must be kept. Remove the whole distal ureter and cuff, with or without LND.
- Non-refluxing reimplant limits seeding; refluxing eases surveillance. This is debated.
- **Boari flap:** a small or irradiated bladder contraindicates it. The base is ≥2 cm wider than the apex, and the width ≥3× the ureteric diameter.
- **Ileal ureter:** 20–25 cm of ileum ≥15 cm from the ileocaecal valve, isoperistaltic. Keep a Foley ≥1 week; do a cystogram and nephrostogram before removal.

Manoeuvre	Length gained	Condition
Psoas hitch	≈5 cm	Mobile bladder; avoid nerve entrapment
Boari flap with hitch	Further 10–15 cm	Adequate capacity
Renal descensus with nephropexy	Up to 8–10 cm	Better on left (longer renal vein)
Ileal ureter / autotransplant	Whole ureter	Autotransplant in skilled hands

4 · Positive cytology & carcinoma in situ (CIS)

A positive cytology with a normal bladder must be traced before any kidney is removed.

ALGORITHM

1. Positive voided cytology: repeat it.
2. CT urogram and bladder biopsies. If the bladder is positive, then treat the bladder.
3. If still positive with a negative bladder (or after bacillus Calmette–Guérin [BCG] "6+1"), then selective upper-tract cytology, retrograde pyelography (RPG), prostatic urethral biopsy (men) and URS.

- Upper-tract CIS is a diagnosis of exclusion: persistently positive selective cytology with normal URS and imaging.
- **No RNU for cytology alone**, because of false positives (infection, stones) and later contralateral disease. Segmental resection fails because of field change.
- AUA/SUO: **6-week BCG induction**, preferably via nephrostomy. Late recurrences can be lethal. RNU if more than surface disease is proven.
- CIS with HG papillary UTUC needs radical or multimodal treatment.
- CIS at the ureteric margin during cystectomy: frozen section; a negative margin is achieved in most, with better survival.

5 · Topical therapy in the upper tract

Drugs drain fast from the upper tract, and pressure or infection can kill.

- Indications: solitary kidney, bilateral UTUC, ESRD risk. Routes: nephrostomy (≤25 cm H_2O), ureteric catheter, stent reflux.
- Always use **low pressure and sterile urine**; exclude infection before every instillation. Sepsis deaths are reported.
- Agents are as in the bladder; evidence is from small uncontrolled series.
- Adjuvant instillation after papillary ablation has **no proven benefit**. BCG is acceptable after complete ablation of HG favourable Ta/T1. Commonest BCG complication: renal granulomas.
- **UGN-101 (Jelmyto)** is mitomycin C (MMC) in a reverse-thermal gel. The Food and Drug Administration (FDA) approved it in 2020 for LG UTUC **not endoscopically manageable**.
- Its toxicity: stricture ≈44% (fewer antegrade), marrow suppression, embryo-fetal toxicity. Complete endoscopic ablation remains the mainstay.

6 · Radical nephroureterectomy

RNU cures only if the whole ureter and cuff come out without spillage.

- **Indications:** non-metastatic high-risk pelvic or proximal ureteric tumour (HG, obstruction, parenchymal invasion), with cuff and LND; or LG with unfavourable features or failed or infeasible endoscopy.
- **Kidney first, cuff last. Clip the ureter early**; control the hilum before manipulation; stay outside Gerota's fascia. **Adrenalectomy is not routine.**
- Never seal a divided ureter with a vessel-sealer alone. Midline is best for nodes and bladder; flank incisions need a second incision for the cuff.
- Remove the **entire intramural ureter and orifice**: a retained stump recurs in **30–75%**, cannot be surveyed and worsens survival. After bowel diversion, excise the ureteroenteric anastomosis with a bowel cuff.
- Extravesical or transvesical (1-cm mucosal cuff) excision is acceptable. Lasso, pluck, stripping and laparoscopic staple are abandoned (spillage, recurrence, positive margins). Close in two

layers; Foley for several days.

- Laparoscopic or robotic: Foley before insufflation; wide perihilar margin; extract **intact in a bag**.
- Ureteric blood supply: anteromedial (upper ⅓), medial (middle ⅓), **lateral** (lower ⅓).
- Relative contraindications: bulky invading tumour, post-chemotherapy LND.

	Open	Laparoscopic	Robotic
Oncology	Reference	Equivalent	Comparable
Morbidity	Highest	Less blood loss, shorter stay	Lowest; fewer positive margins
Caveat	—	Possibly more bladder recurrence; less favoured for T3/T4	Longest; more LND done

> **KEY POINT**
> Principle beats approach: port-site seeding and carcinomatosis follow spillage, not the approach.

Lymphadenectomy

- P4 Role of neoadjuvant therapy and Lymphadenectomy for muscle invasive bladder cancer and upper tract urothelial cell cancer? — *UTUC templates (right interaortocaval), ≥8 nodes, chylous leak; neoadjuvant rationale, EA8141, AUA/SUO CrCl <50, POUT*

Not in the paper answer:

- Caudal limits, T2-to-T3 cliff, surgeon sets yield, pN0 > pNx > pN+ (below); cN+ chemotherapy first (§7).
- LND has value in **T2–T4**, especially pelvis and proximal ureter. The surgeon, not the approach, sets the yield. Outcome: pN0 better than pNx, better than pN+.
- Caudal limits: renal pelvis to the inferior mesenteric artery; upper ⅔ ureter to the aortic bifurcation; lower ⅓ to the pelvis.
- The prognostic cliff is **T2 to T3**. HG behaves like T3, and N+ disease treated by surgery alone behaves like metastatic disease. Renal function loss is RNU's main cost.

7 · Perioperative adjuncts

Bladder recurrence and lost cisplatin eligibility are the two predictable problems after RNU.

- Bladder tumours follow in 30–50% after RNU or endoscopy, mostly within 2 yr. **AUA/SUO strongly recommend** a single dose after RNU (optional after endoscopy).
- Post-op (day 3–7): cystogram first, then MMC or gemcitabine, dwell 1–2 h. **Intraoperative** option: instil before surgery, clamp 1–2 h, drain once the ureter is clipped; no cystogram needed, and drug is present when seeding occurs.
- **Renal function is the hinge**, so give platinum **before** nephrectomy. Neoadjuvant is favoured if post-op creatinine clearance (CrCl) is expected **<50** mL/min.
- EA8141: CrCl >50 received accelerated methotrexate, vinblastine, doxorubicin and cisplatin (MVAC) ×4.
- Adjuvant (POUT): gemcitabine–cisplatin (CrCl >50) or gemcitabine–carboplatin (30–49); cisplatin preferred. cN+ disease: chemotherapy, then consolidative surgery.

8 · Complications

Most are preventable by technique.

Procedure	Complication	Management / prevention
URS	Perforation; stricture	Stent; pre-stent tight ureter; no circumferential fulguration
Percutaneous	Bleeding, irrigant absorption, perforation, secondary ureteropelvic junction obstruction	Bipolar resection; avoid major vessels; nephrostomy
Topical therapy	Sepsis, absorption, stricture	Sterile urine, low pressure
Cuff closure	Urine leak	Two-layer closure; catheter; cystogram
RNU	Renal insufficiency	Nephrology referral

9 · Follow-up

Recurrence is common and late, so surveillance is lifelong and risk-based.

- Every visit: **cystoscopy**, cytology, metabolic panel; after KSM add ipsilateral URS. Do not routinely scope a normal contralateral tract.
- CT or MR urography (RPG if no contrast); filling defects need URS. Add chest imaging and smoking cessation.

Group	Cystoscopy	Upper-tract endoscopy	Imaging
KSM low risk	3 mo, then 6–9-monthly ×2 yr, then yearly	3 mo, 6 mo, 1 yr	6–9-monthly ×2 yr, yearly to 5 yr
KSM high risk	3 mo, then 3–6-monthly ×3 yr, then yearly	3 mo, 6 mo, 1 yr	3–6-monthly ×3 yr, yearly to 5 yr
RNU <pT2N0	With cytology at 3 mo, then by grade	—	CT/MRI by 6 mo, yearly ≥5 yr
RNU ≥pT2	3 mo, 3–6-monthly ×3 yr, yearly	—	Abdomen/pelvis 3–6-monthly ×2 yr; chest CT 6–12-monthly ×5 yr

10 · Landmark trials

These trials make single-dose instillation and adjuvant platinum standard.

Trial	Design	Result	Take-home
ODMIT-C	Randomised controlled trial (RCT), MMC 40 mg single dose at catheter removal after RNU	Bladder recurrence 28% vs 16% at 1 yr	Single-dose instillation standard
Pirarubicin trial (Ito)	RCT, single pirarubicin dose within 5 days of RNU	Bladder recurrence 42% vs 17% at 2 yr	Confirms single post-RNU instillation
Laparoscopic vs open RNU	Only RCT; small	Equal 5-yr survival; less blood loss	Minimally invasive acceptable
UGN-101 phase III	Single-arm, LG UTUC	Complete response 59%; 56% disease-free at 12 mo	Gel chemoablation option

Trial	Design	Result	Take-home
POUT	Phase III adjuvant platinum vs surveillance, pT2–4/N+	Stopped early; disease-free survival (DFS) hazard ratio **0.45**; 3-yr DFS 71% vs 46%	Adjuvant platinum standard after RNU for ≥pT2/N+

141 Contemporary Open Surgery of the Kidney

1 · Role & preoperative work-up

Open surgery is kept for the hardest cases, so find the risks beforehand.

- Open surgery offers **exposure and early great-vessel control**. It suits renal cell carcinoma (RCC) with inferior vena cava (IVC) thrombus, cytoreduction, complex tumours and a hostile abdomen (minimally invasive surgery: *Ch 142*).
- **Renal function** decides partial (PN) versus radical nephrectomy (RN). Estimate GFR (eGFR) with the **CKD-EPI 2021 race-free** equation, which beats MDRD; cystatin C detects mild glomerular filtration rate (GFR) loss.
- If GFR is **<60** or there is significant proteinuria, ask for a perioperative nephrology consult. Optimise diabetes and hypertension even if GFR is ≥60.
- Advanced RCC: blood count and metabolic panel (MSKCC and IMDC risk models), liver tests and coagulation (**Stauffer syndrome**, paraneoplastic hepatic dysfunction), cardiopulmonary fitness.
- Contrast is generally safe if GFR is **>30**. If thrombus is suspected, MR or CT venography must be **≤1 month old**, because thrombus propagates rapidly.
- Prophylaxis: cefazolin (or clindamycin) ≤1 h pre-incision; no routine bowel preparation; pneumatic compression and subcutaneous heparin 5000 IU; stop smoking 4–6 wk.
- **Renal artery embolisation** lets the vein be tied before the artery when the hilum is hidden. It is no longer routine: it causes postinfarction syndrome (pain, fever ≤72 h), may *increase* blood loss, and makes surgery harder if delayed >48 h.
- If used, embolise the evening before or morning of surgery and confirm inflow is occluded.
- Use atraumatic vascular clamps (bulldog, Satinsky), never a Kocher, on vessels. Use the argon beam for liver tears.

2 · Surgical anatomy & approaches

Most disasters come from misread anatomy.

- At risk: **right**, liver, duodenum, colon; **left**, spleen, pancreatic tail, colon, superior mesenteric artery (SMA).
- The **right renal artery** lies posterior-inferior to the renal vein, behind the IVC. It can be reached in the **interaortocaval space** by retracting the left renal vein (LRV) upward.
- The **SMA** leaves the front of the aorta **1–2 cm above LRV** and veers left, so it can be mistaken for the left renal artery.
- The right gonadal and adrenal veins drain into the IVC; the left ones drain into the LRV.
- Anterior exposure means reflecting the colon off Gerota's fascia along the **white line of Toldt**. On the right, add a **Kocher** manoeuvre without cautery, because duodenal injury is devastating. On the left, divide the splenorenal ligament.
- **Cattell-Braasch:** Toldt's incision round the caecum, Kocher, and the mesenteric root freed to Treitz; the midgut pivots on the SMA, exposing aorta and IVC.
- **Infracolic** (trauma, bulky hilum): below Treitz to the aorta and LRV. Partial **Mattox** (left): spleen and pancreas rotated medially together.

Incision	Best for	Limitations
Midline	Trauma; advanced tumour or IVC thrombus (early aortic access); bilateral	Bowel mobilisation needed
Anterior subcostal	Unilateral transperitoneal	Muscle division
Chevron	Renovascular; RN with thrombectomy; both pedicles	Extensive muscle division
Subcostal flank (below 12th rib)	Lower pole, pelviureteric junction, proximal ureter	Poor hilar access; not for PN
Supracostal (11th/12th)	RN or PN; 8-cm mini-flank above the 11th	Pleura
Transcostal (11th, rib resected)	Retroperitoneum at an old rib tip	Rib division
Thoracoabdominal, dorsal lumbotomy	Largely historical; thoracoabdominal is still an option for level III–IV thrombus	

- **Supine:** excess hyperlordosis causes back pain, so avoid it in spinal stenosis.
- **Flank:** 12th rib tip over the kidney bar, axillary roll. It lowers venous return and lung function, so it is poor in cardiopulmonary disease or severe kyphosis. Pad against rhabdomyolysis.
- **Supracostal** (popularised by Turner-Warwick): incise on the upper rib border, open the lumbodorsal fascia at the rib tip, push the pleura up, and divide the intercostals close to the **upper border of the rib** (spares the bundle).
- Spare the **subcostal nerve**, which runs between the internal oblique and transversus.

3 · Simple nephrectomy

For benign disease, stay close to the kidney.

- Stay **within Gerota's fascia** for a benign nonfunctioning kidney (obstruction, infection, stones, reflux, polycystic, dysplasia). A functioning kidney is rarely removed (pain, bleeding, renovascular hypertension). Stay on the capsule and spare the adrenal.
- Benign PN suits focal atrophy, a duplex pole, an infected diverticulum or a symptomatic angiomyolipoma.

4 · Radical nephrectomy

RN is for tumours PN cannot treat; the key skill is hilar control.

- Robson (1969) removed the kidney, Gerota's fascia, adrenal and nodes from crus to bifurcation. Now the **normal adrenal is spared** and lymph node dissection (LND) is selective.
- **Indications:** tumours not amenable to PN, such as a nonfunctioning kidney, tumour replacing the kidney, nodes, cytoreduction or venous thrombus.
- Staging: abdominal CT or MRI and **chest CT**; bone scan only for symptoms, fracture, suspicious imaging or raised bone alkaline phosphatase; no routine brain imaging.
- Biopsy only if it changes the plan: infiltrative mass (urothelial, lymphoma, metastasis), before systemic therapy, bilateral, trials.
- T4 invasion means altered enhancement of the organ, not just contact. The liver is rarely truly invaded (depth usually <1 cm).

ALGORITHM

1. **Right:** survey the abdomen; open Toldt's line down to the iliac vessels; take down the hepatic flexure; Kocher; expose the IVC.
2. Loop and palpate the renal vein; find the artery behind it; tie **artery, then vein**; remove the kidney outside Gerota's.
3. **Difficult hilum:** ligate the artery in the interaortocaval space, so the kidney decompresses. Check the now flaccid vein and double-ligate it (0 silk and 3-0 Prolene), then divide the artery laterally.
4. Alternative: start at the lower pole, divide the ureter and gonadal vein, and lift the kidney from posterior to anterior to show the artery.
5. **Left:** mobilise the splenic flexure down to the iliacs; divide the renocolic ligament (pad the pancreas).
6. Follow the aorta, or gonadal vein upward, to the LRV. Tie the artery above and behind it (0 silk and 3-0 Prolene), confirm occlusion, then the vein.

KEY POINT

Why artery first? Tying the vein first leaves the kidney engorged, so collaterals bleed. Arterial ligation shrinks the kidney and softens the vein, which can then be checked for thrombus. A persisting pulse means an accessory artery, incomplete control, or the SMA has been tied.

- **En bloc pedicle** ligation (with hilar nodes): arteriovenous fistula (AVF) is a theoretical, very rare risk.
- **Lumbar veins:** use silk ties and **no clips on the IVC side**, because clips block emergency caval clamping.
- **Lost pedicle control:** tell the anaesthetist, compress, use two suckers, **never clamp blindly**.
- **Adrenalectomy** only for an upper-pole tumour with a lost plane, an adrenal mass or diffuse involvement. Metastasis occurs in **<5%**, and CT or MRI misses 20–25%.
- Contralateral metastasis is as common as ipsilateral, so prophylactic removal risks adrenal insufficiency without benefit.
- **LND:** nodes are positive in ≈5%. The EORTC RCT (772 patients) showed no overall survival benefit, but it included many low-stage tumours.
- **Do LND if nodes are enlarged**: a short axis **≥1.5 cm ≈ 50% positive**. Consider it in high risk (cytoreduction, >10 cm, grade ≥3, sarcomatoid, thrombus, T4).
- Templates: right, paracaval, precaval, retrocaval and interaortocaval; left, para- and retroaortic; from crus to bifurcation. **Clip lymphatics above the renal vein** to prevent chylous ascites from the cisterna chyli.
- **Closure:** fill with saline and give a big breath; bubbles mean a pleural tear. Suture it over a 14-Fr catheter and withdraw on a held breath. Take a **chest X-ray in all**.

5 · IVC tumour thrombectomy

Surgery is the only cure, and the thrombus level decides the plan.

- RCC is the commonest cause (**5–10%** of RCC); others are Wilms tumour and adrenocortical carcinoma. Untreated, most die within 1 yr. Surgery (RN and LND, with or without systemic therapy) is the backbone.

- **Tumour thrombus** differs from **bland thrombus** (clot below the obstruction), and they are managed differently. **MR venography** best shows extent, wall invasion and bland thrombus.
- Screen for pulmonary embolism (PE) and deep vein thrombosis; anticoagulate for venous thromboembolism, bland thrombus or IVC occlusion.
- **No suprarenal filters**, because thrombus grows onto the device. Embolise only for unexcisable hepatic-vein thrombus or a bleeding kidney.
- Use **transoesophageal echo (TEE)** for levels II–IV. Involve a vascular or liver surgeon for II–III and a **cardiothoracic surgeon for III–IV**.

Mayo level	Upper extent	Management
0	Renal vein	RN (commonest)
I	IVC **≤2 cm** above ostium	Milk back, Satinsky partial clamp, ostial cavotomy
II	>2 cm, below hepatic veins	Full IVC control; infrahepatic cavotomy
III	At or above major hepatic veins, below diaphragm	Liver mobilisation with or without Pringle; occlusive thrombus: suprahepatic clamp; nonocclusive: venovenous bypass (VVB)
IV	Above diaphragm	Sternotomy, cardiopulmonary bypass (CPB) with deep hypothermic circulatory arrest (DHCA), atriotomy

Bland group	Location	Action
A	None	None
B	Iliac veins	Infrarenal filter
C	Infrarenal IVC, separate	Interrupt (clip or staple); do not extract
D	Mixed with tumour	Resect infrarenal IVC; keep lumbars as collaterals

ALGORITHM

1. **Level I (right):** control the IVC before handling the tumour; ligate the right artery interaortocavally.
2. Loop the LRV and the suprarenal and infrarenal IVC with **Rummel tourniquets**.
3. Milk the thrombus toward the ostium; place a Satinsky round the ostium with the thrombus inside the jaws.
4. Do an ostial cavotomy keeping ≥half the IVC width, and extract.
5. **Level I–II:** ligate the renal artery early. Tighten tourniquets in order: **infrarenal IVC, contralateral renal vein, suprarenal IVC**.
6. Open the cava, extract, flush with heparinised saline and inspect the wall (resect suspicious intima). Close with 4-0 Prolene.
7. Release the **caudal clamp first**, flush 5–10 mL on a held breath, then release the cranial clamp.

- **Left tumour (level II)** is harder, because the IVC is reached from the right and both colons are mobilised. Tie the left artery first, then Kocher, **split-and-roll** the IVC and excise the LRV ostium en bloc.
- Do not clamp the right renal artery (no venous collaterals); release the right renal vein clamp intermittently. No routine heparin.
- **Level III:** keep bypass on standby and mobilise the liver as in transplantation. The **major hepatic veins are never divided**. Tying the accessory (caudate lobe) veins gives 2–3 cm

more IVC.

- Level III is defined by the major, not accessory, hepatic veins. **Budd–Chiari syndrome is a contraindication**.
- The **Pringle** manoeuvre (clamping the portal vein, hepatic artery and bile duct) is needed if clamping above the hepatic veins. Limit it to ≤60 min, ideally **≤20 min**. Pringle with clamps above and below the liver gives total hepatic vascular occlusion.
- **Level III sequence:** clamp infrarenal IVC, contralateral renal vein, porta hepatis, then suprahepatic IVC; open and extract (Fogarty or 20-Fr Foley for unreachable thrombus). Unscrapable wall invasion needs IVC resection.
- **Clamp trial** for 2–5 min: preload falls by 60% and cardiac output (CO) by 40%. If CO falls by >50% or mean arterial pressure (MAP) by >30%, use bypass or a supracoeliac aortic clamp.
- Occlusive thrombus tolerates clamping (azygos collaterals); free-floating does not. Bypass also for cardiac or hepatic dysfunction, a poor contralateral kidney, portal hypertension or uncontrolled bleeding.
- Incomplete tumour removal is universally fatal, so never compromise IVC control to save time.

	VVB	CPB with DHCA
Access	Femoral to jugular, percutaneous	Sternotomy; cool to 16–18 °C and drain 95% of blood
Heparin	Partial	Full; coagulopathy
Drawback	Lumbar and intercostal back-bleeding	Cardiac and neurological events
Use	Least invasive; not with atrial or iliac bland thrombus, Budd–Chiari	Level IV; bloodless field; ≤60 min (<40 better)

- **VVB:** place cannula tips far from the thrombus, because dislodgement causes massive PE. **DHCA** is used when the thrombus cannot be milked below an intrapericardial clamp.
- P3 Vascular grafts and vascular surgery in urology — *caval repair: ≥50% lumen primary closure, else pericardial patch (Leibovich–Bower width), circumferential: PTFE 16–20 mm + lifelong aspirin 81 mg; left tumour needs right-kidney drainage*

Not in the paper answer:

- No suprarenal filters; bland groups A–D; clamp-trial limits; Pringle ideally ≤20 min; DHCA ≤60 min; suprarenal ligation for right tumours (all above); no cell saver (§7).
- Patch: 5-0 Prolene, small regular bites. Suprarenal IVC ligation with a right tumour is acceptable if the LRV is tied caval to its tributaries (collaterals).

> KEY POINT
> **"Shortcuts" that become "longcuts":** opening the IVC before all inflow is controlled; side-biting a level I; clips on the IVC side; incomplete arterial control; forceful milking (embolus, tear); clamping before volume loading or skipping the test clamp.

6 · Partial nephrectomy

PN balances a clear margin against nephrons kept.

- P2 Nephron-sparing surgery / partial nephrectomy in RCC: role, indications and approaches — *PN standard for T1; imperative/relative/elective; relative contraindications (<20% nephron mass, pedicle encasement, thrombus, cT4, nodes); hereditary observe to 3 cm except HLRCC*

Not in the paper answer:

- Collecting-system invasion as a contraindication; hyperfiltration only if nephron mass falls >80%; multifocality rates; RCT table; 5–10 mm rim; no permanent sutures (below); urine-leak and pseudoaneurysm algorithms (§7).
- PN is size-independent if it is safe. Collecting-system invasion is also a relative contraindication.
- **Hyperfiltration** (fewer nephrons each filter more, causing focal segmental glomerulosclerosis [FSGS], proteinuria and chronic kidney disease) is rare unless nephron mass falls by >80%.
- Multifocality: clear cell ≈2%, papillary ≈10%, so inspect the whole kidney with intraoperative ultrasound (IOUS).
- In hereditary RCC, preserve and reapproximate the perirenal fat and Gerota's fascia, which eases later surgery.

Question	Evidence
Warm ischaemia	Aim **<30 min**; within limits, preserved parenchyma dominates
Cold vs none	Randomised controlled trial (RCT): no difference
Mannitol	RCT: no benefit
On- vs off-clamp	2 RCTs: no difference
3D models	RCT: less ischaemia, blood loss and stay

> **KEY POINT**
> PN trades nephrons lost to the margin against nephrons injured by ischaemia. Within accepted limits, the **volume of parenchyma preserved** matters more than clamp minutes. Hypothermia and mannitol are optional (their role in solitary kidney or CKD is unclear).

- Always **control the hilum first**, even if not clamping, so a clamp can go on at once.
- Options: manual compression, parenchymal clamp, selective branch clamp, whole pedicle, early unclamping or off-clamp. Leaving the vein open has no proven benefit.
- **Enucleation** follows the pseudocapsule plane with minimal normal tissue. **Resection** takes a 5–10 mm rim, is usually clamped and transects more small vessels. Send the deep margin for frozen section if needed.
- **Renorrhaphy:** inner layer, 4-0 absorbable to vessels and running collecting-system closure (no permanent sutures, which become a stone nidus); outer layer, capsular sutures over a bolster. Drain if the collecting system was entered.

7 · Complications & management

Each injury has a set response.

Injury	Management
SMA	Injured when mistaken for the left renal artery or by Cattell-Braasch traction. Potentially lethal, so call a vascular surgeon for primary repair or graft
Inferior mesenteric artery or vein (IMA/IMV)	Ligation safe if the marginal artery of Drummond is patent
Superior mesenteric vein (SMV)	Ligation causes bowel ischaemia, splanchnic hypertension and hypotension. **Never ligate**; repair with 6-0 Prolene venorrhaphy. If transected, reanastomose or use autologous vein; leave the abdomen open

Injury	Management
Duodenum	Haematoma narrowing the lumen: incise serosa and muscularis, drain. Tear: debride, 2-layer closure, omentum, drain, nasogastric tube
Pancreas	Glue and drain (check amylase); ductal injury: general surgeon; late collection: percutaneous drain, nil by mouth, octreotide
Pleura	Purse-string over a 14-Fr catheter, evacuate on a positive-pressure breath; large tear: chest tube

- **Thrombectomy:** air embolism is prevented by releasing the caudal clamp first and letting ≈5 mL blood escape. For PE with collapse, do a pulmonary arteriotomy.
- Bleeding despite caval control means an untied lumbar vein. Use **no cell saver**, because it spreads tumour. Liver enzymes peak at 2–3 days.
- **Urine leak** (large, endophytic, hilar tumours) mostly appears within 72 h. If output is low, remove the drain by 48 h.
- If output is >50 mL/24 h, check drain creatinine; a value above serum means urine. A urinoma persisting >48 h, or symptoms >24 h, needs a retrograde double-J stent and Foley. Most resolve in 4–6 wk.
- **Pseudoaneurysm** presents with flank pain, haematuria or shock. If stable, use bed rest and serial haemoglobin.
- If significant, do **angiography** with **selective embolisation**, because CT angiography can miss it and cannot treat it. Re-explore if refractory.
- **AVF** causes bruit, haematuria, heart failure or steal ischaemia; diagnose by Doppler, CT or MR angiography. Observe, then coil embolisation, then surgery if complex.
- **Renal insufficiency** is usually mild and transient (solitary kidney, large resection, long ischaemia). Exclude obstruction, drug toxicity and thrombosis.

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1 · Principles & patient selection

Fitness here means fitness for the gas.

- First laparoscopic nephrectomy: **Clayman, 1991** MCQ; first laparoscopic partial nephrectomy (PN): Winfield, 1993.
- Minimally invasive surgery (MIS) gives less pain, blood loss and convalescence than open, with **equivalent long-term outcomes** and the same indications.
- **General anaesthesia is mandatory.** The CO2 pneumoperitoneum impairs ventilation, venous return and cardiac output, so risk is high in chronic obstructive pulmonary disease (COPD), intracranial pathology and cardiac disease.
- Severe COPD with hypercarbia: lower the pressure, then helium, then convert. AirSeal reduces CO2 load.
- Robot: **docking and undocking delay emergency access**, and the console surgeon may miss the patient slipping, so communicate closely.
- In obesity, **visceral adipose tissue**, not body mass index (BMI), predicts difficulty. Age is no barrier.
- Prior surgery changes trocar sites and the choice of transperitoneal or retroperitoneal route, but it is **not a contraindication**.

> **KEY POINT**
> The patient must tolerate the **pneumoperitoneum**, not the renal operation: cardiopulmonary reserve, not age or BMI, decides fitness.

2 · Access & port placement

Tumour position and the abdomen choose the route.

Routes: transperitoneal (TP, most used), retroperitoneal (RP), hand-assisted laparoscopy (HAL), laparoendoscopic single-site surgery (LESS) and natural orifice surgery (NOTES).

	Transperitoneal	Retroperitoneal
Space / landmarks	Large; familiar	Limited, poor triangulation; less familiar
Bowel	Colon reflection; adhesiolysis	Peritoneum not entered
Hilum	Reached after colon reflection	Direct, especially the posterior aspect
Preferred for	Most surgery; suturing; anterior tumours	Posterior tumours, pyeloplasty, cysts, biopsy; dense adhesions, peritoneal dialysis, pregnancy, morbid obesity

- **TP:** modified flank position at 30–45°, with 3–5 trocars (umbilical camera, subxiphoid-midline, and a 12 mm stapler port in the anterior axillary line). In obesity, shift the ports laterally.
- **Robotic** arms are spaced **8 cm (Si) / 6 cm (Xi)**, with a 12 mm assistant port. Ports form a **broad-based triangle** with the camera at the apex.
- **RP:** full flank, table flexed; incise in the posterior axillary line between 12th rib and iliac crest. Create the space by finger (psoas, lower pole) or balloon (600–800 mL), then insert a

blunt-tip cuffed trocar.

> **KEY POINT**
> RP pitfalls: a balloon inflated within muscle causes a flank hernia; too anterior injures the peritoneum or colon; too posterior makes the quadratus or psoas bleed. Bail-out: open the peritoneum and continue transperitoneally.

- **HAL** gives tactile feedback and manual retraction. It helps novices, donors and scarred kidneys (xanthogranulomatous pyelonephritis, XGP).
- Drawbacks: forearm paraesthesia, wound infection or hernia; bag the specimen (hand-port metastases). In an emergency, a hand through the extended port compresses the bleeding vessel.
- **LESS and the single-port (SP) robot** improve cosmesis and pain but lose triangulation. Outcomes are unchanged, but SP partial nephrectomy has longer warm ischaemia.

3 · Nephrectomy — operative steps

The steps copy open surgery.

> **ALGORITHM**
> 1. Reflect colon; find ureter and gonadal vessels on the psoas.
> 2. Lift kidney to expose the hilum; artery, then vein.
> 3. Upper pole (adrenal spared or taken); ureter and lateral attachments; bag; extract.

- **Colon:** keep the **lateral attachments** until late, or the kidney rolls over the hilum. The plane lies between bright-yellow mesenteric and pale Gerota's fat; close mesocolic defects. Right side: cold dissection near the duodenum, **Kocher**.
- The **ureter** lies just behind the gonadal vessels and is recognised by peristalsis. Use it as a handle, and stay above the psoas fascia to avoid thigh numbness.
- **Hilum:** divide the **artery first, then the vein**, with an endo-GIA stapler. Staple the left renal vein lateral to the adrenal vein. **Never leave clips in the staple line.**
- **En bloc hilar stapling** is safe (no arteriovenous fistula, AVF) and saves ~45 min.
- **Upper pole:** to spare the adrenal, open Gerota's above the hilum and peel fat off the pole. With extreme fibrosis, go subcapsular once vessels are controlled.
- **Extraction:** intact in a bag, or morcellated in an impermeable **LapSac**, which keeps type and grade but loses size and stage.

4 · Simple vs radical nephrectomy

Inflamed benign kidneys are often harder than cancers.

	Simple	Radical
Gerota's fascia and fat	May be left	Removed intact with specimen
Pathology	Benign, often inflammatory	Renal cell carcinoma (RCC)
Difficulty / conversion	Can exceed radical; higher	Planes preserved; lower

- **Simple:** chronic pyelonephritis or XGP, obstructive or reflux nephropathy, tuberculosis, multicystic dysplastic kidney, renovascular hypertension, acquired cystic disease, symptomatic polycystic kidney disease.

- Hardest after pyonephrosis, nephrostomy or stones: use hand assistance or subcapsular nephrectomy.
- **Radical nephrectomy (RN):** MIS equals open oncologically at 5 and 10 yr. In RCTs of RP versus TP, the only consistent RP advantage is **shorter operative time**.
- Tumours >10 cm are feasible (landmarks lost; use ultrasound, extra ports). Conversion is higher laparoscopic than robotic.
- **Lymph node dissection (LND)** has staging value only, with no overall survival benefit. The American Urological Association (AUA) advises removing clinically positive nodes.
- **Inferior vena cava (IVC) thrombus** (experts): early arterial ligation may retract it; milk back, then staple or Satinsky the cuff. Preop embolisation helps robotic (not open) surgery (*Ch 141*).
- **Cytoreductive** nephrectomy by laparoscopy allows quicker recovery and earlier systemic therapy.

> **KEY POINT**
> Access changes morbidity and recovery, never oncology: early arterial control, en bloc removal and intact bagged extraction carry over from open surgery.

5 · Partial nephrectomy

PN is the default where feasible.

- P2 P1 Nephron-sparing surgery / partial nephrectomy in RCC; Role of renal nephrometry in management of RCC — *PN default where feasible; RENAL score predicts complexity, WIT, conversion; enucleation; TP anterior vs RP posterior*

Not in the paper answer:

- >5 cm pT3a caveat; heminephrectomy ≥30%; collecting system entered ~60% (always drain); single-layer renorrhaphy; clamping verdicts; margins, port-site recurrence and volume plateau (all below).
- In cT1b/T2 tumours, PN gives better estimated GFR (eGFR) with equal oncology. **Caveat: tumours >5 cm upstaged to pT3a recur more after PN**, so for large tumours the case for PN is renal function.
- A **hilar** tumour touches the renal artery or vein; consider enucleation and use superficial renorrhaphy bites.
- Multifocal: selective clamping guided by 3D reconstruction. Solitary kidney: robotic PN gives short warm ischaemia time (WIT) and rare conversion.
- **Heminephrectomy** means removing ≥30% of the parenchyma.

> **ALGORITHM**
> 1. Intraoperative ultrasound (size, depth); open Gerota's away from the lesion; score the capsule.
> 2. Clamp the hilum (Satinsky en bloc or bulldogs); cold-scissor excision.
> 3. Close vessels and collecting system; renorrhaphy; unclamp; drain.

- **Trifecta:** negative margin, no major complication within 30 days, and **WIT ≤25 min**.
- **Enucleation** gives less blood and function loss but more positive margins, with the same survival. It suits familial, multifocal, solitary-kidney and severe chronic kidney disease (CKD) cases.

- **Renorrhaphy:** the sliding-clip (Lapra-Ty) and barbed sutures shorten WIT. **Single-layer** closure gives better function and fewer pseudoaneurysms. Sealants show no randomised benefit.
- The **collecting system** is entered in **~60%**. Repair it with 2-0/3-0 polyglactin and **always drain**; a routine ureteric catheter does not prevent fistula.
- **Ischaemia:** the historical warm limit is **30 min**, but **preserved parenchymal volume predicts function better than WIT**. Use cold ischaemia for a solitary kidney, CKD or a long expected clamp time.

Hilar strategy	Best for	Verdict
Artery-only vs artery and vein	Any	Equivalent
Early unclamping	Any	Possibly fewer pseudoaneurysms
Selective / super-selective (near-infrared fluorescence, duplex)	Mapped segmental supply	Randomised: no 6-month benefit
Off-clamp (zero ischaemia)	Small, exophytic	Better early eGFR; no 2-yr difference; more blood loss
Parenchymal compression	Peripheral	Feasible, safe
Hypothermia	Solitary kidney, long WIT	Benefit unproven

KEY POINT
No clamping strategy has beaten standard clamping on long-term function: what is kept matters more than how long it is clamped.

- Robotic versus laparoscopic PN: fewer conversions and shorter WIT. Robotic versus open: less blood loss and fewer complications. Morbidity plateaus at 18–20 robotic PN/year.
- **Positive margins (0–7%)** matter only in high-risk tumours (pT2–3, grade 3–4), so the answer is **surveillance, not reoperation**.
- **Port-site recurrence is ~0.1%**. It follows spillage and technique, not the gas, so use an impermeable bag.
- Local recurrence after RN is ~2% and is treated by resection with systemic therapy. Salvage after PN or ablation often ends in RN, because the planes are lost.

6 · Other renal procedures

Know the indication and one key step for each.

Procedure	Indication	Key points
Cyst decortication	Pain, infection, obstruction, suspected malignancy	Percutaneous aspiration first proves the cyst causes the pain; excise the wall, biopsy, fulgurate the base
Laparoscopic renal biopsy	Failed percutaneous biopsy, bleeding diathesis, solitary kidney, morbid obesity	RP, two ports, lower-pole cortex
Nephropexy	Descent >5 cm or 2 vertebrae from supine to erect, with pain, reduced flow or obstruction	Usually right side, slim young women; 20% asymptomatic. Fix the capsule to psoas or quadratus fascia

Procedure	Indication	Key points
Calyceal diverticulectomy	Symptomatic; endoscopy failed	Suture the neck, obliterate the cavity with fat
Nephrolysis	Chyluria, loin pain–haematuria	Lateralise by cystoscopy; hilar lymphatic ligation with or without omental wrap
Ablation	≤3 cm, comorbid, solitary kidney, hereditary RCC	Percutaneous preferred. Cryo: two freeze–thaw cycles, iceball 0.5–1 cm beyond tumour. Radiofrequency ablation (RFA): 60–100°C (>105°C vaporises), 1 cm margin, no real-time imaging; more retreatment

7 · Complications & management

Dangerous complications happen out of view, so seek them.

- P1 Pneumoperitoneum physiology, complications of retroperitoneoscopy and laparoscopic donor nephrectomy — *vascular injury commonest (<1/1000, fatal 8–17%); venous bleed control; gas embolism (left side down, head down, aspirate); pneumothorax; thermal bowel injury, CT with oral contrast; rhabdomyolysis*

Not in the paper answer:

- Overall rate, learning curve, device-failure ranking, left-sided SMA trap, end-of-case pressure drop, day-12 haematuria, urine leak, bowel-injury presentation, pneumothorax and neuromuscular rates (all below).
- The overall rate is **~15%**, and PN > RN > simple. The learning curve is ~20 nephrectomies. Robotic PN halves open conversion. Age is not a major risk; conversions cluster in infective disease.
- Vascular injury happens at **access** or at the hilum. Device failure ranks staplers > non-locking clips > locking clips, mostly from operator error.
- **In left RN, the superior mesenteric artery (SMA) can be mistaken for the renal artery**, which leads to bowel necrosis.

> **ALGORITHM**
>
> 1. Major bleed: locate; temporary control (grasper or pressure); inform the anaesthetist (blood, open set).
> 2. Clear the field: raise pneumoperitoneum or add a trocar; then definitive clip or suture.
> 3. If that fails, then convert and call a vascular surgeon.

- Hilar injury comes from dividing before identifying, sharp hilar dissection, or a stapler not covering the vessel.
- Venous ooze: pressure. Brisk arterial bleed: hand through a lower midline incision. **Never clip or suture blindly.**
- At the end of the case, **lower the insufflation pressure** to unmask venous bleeding (bed, adrenal, mesentery, gonadal vessels, ureteric stump).
- Post-op haemorrhage: CT; arterial source: angioembolisation.
- **Haematuria at ~day 12 after PN** suggests a pseudoaneurysm or AVF; treat by selective embolisation.
- **Urine leak** after PN (0.8–10%; large or hilar tumour, collecting-system repair, long WIT): percutaneous drain; if persistent, stent and catheter or nephrostomy.

- **Bowel injury** (<1%, mortality ~3%) is thermal in ≈ half and occurs at access in 32%. It presents with **pain at a single trocar site**, distension, diarrhoea and a **normal or low white count**.
- Stable: laparoscopic repair; unstable or gross spillage: laparotomy.
- **Solid organs:** spleen (left traction); pancreas, where injury is insidious, so inspect at the end; gallbladder needs cholecystectomy.
- **Diaphragm:** sudden rise in airway pressure; purse-string and chest X-ray. Unrecognised pneumothorax ~0.3%.
- **Neuromuscular** (2.7%): flank rhabdomyolysis needs hydration; an axillary roll protects the brachial plexus. Venous thromboembolism is lower than open.

> **KEY POINT**
> MIS complications are the open ones in altered form and timing: the lethal ones (access vessel injury, thermal bowel injury) happen out of view and present late or atypically.

8 · Laparoscopic live donor nephrectomy †

The donor is healthy, so donor safety comes first.

- P4 P1 Laparoscopic donor nephrectomy — *left preferred (longer vein), donor keeps better kidney; gonadal–ureter packet; artery then vein; no Hem-o-lok on donor artery; mortality 3/10 000; lifetime ESRD risk*

Not in the paper answer:

- † Split-function renogram if kidney sizes differ; chylous ascites; lifelong annual checks (below).
- † Complications include chylous ascites. Donors need lifelong annual blood pressure and renal function checks.

143 Focal Therapy for Renal Tumours

1 · Rationale & indications

Why this matters: ablation suits patients whom surgery harms more than the tumour.

- Small renal cell carcinoma (RCC) rarely kills; chronic kidney disease (CKD) and comorbidity do. So minimise competing-risk harm (*see Ch 137*).
- **Partial nephrectomy (PN) is the gold standard** for cT1a. Thermal ablation (TA) is the alternative: cryoablation (CA), radiofrequency ablation (RFA) or microwave ablation (MWA).
- **Indications:** high surgical risk, solitary kidney, bilateral tumours, or hereditary syndromes (von Hippel–Lindau, VHL) needing repeat treatment. Recovery is faster, complications fewer, and renal function equal or better than PN.

KEY POINT

Trap: TA is never the default for a fit patient.

Guideline	Position on TA
American Urological Association (AUA) 2021	Consider for cT1a **≤3 cm**, especially comorbid; PN prioritised; percutaneous preferred
European Association of Urology (EAU) 2022	TA not routinely **>3 cm**; CA not **>4 cm**; for frail/comorbid
National Comprehensive Cancer Network (NCCN) 2022	**Only one allowing cT1b** (non-surgical candidates); may need multiple treatments
American Society of Clinical Oncology (ASCO) 2017	Option if complete ablation achievable

2 · Selection & work-up

Why this matters: the tumour stays in place, so biopsy gives the only histology.

ALGORITHM

1. Counsel: age, comorbidity, CKD risk, size and location.
2. Biopsy, then re-counsel with histology.
3. Choose PN, surveillance or percutaneous ablation.

- **The AUA advises biopsy of all**, ideally at a **separate earlier session** (repeatable; fewer benign masses ablated). Routine post-ablation biopsy adds nothing; biopsy only a new enhancing lesion.
- Treat urinary tract infection (UTI) first; colonised urine (ileal conduit) risks abscess.
- Complication predictors: RENAL score (low 4–6, moderate 7–9, high 10–12), American Society of Anesthesiologists (ASA) score >3 (about 3×), $(MC)^2$ (CA only). Day-case suits ≤3 cm, low complexity, no anticoagulation.

3 · Mechanisms

Why this matters: every technical rule aims to carry a lethal temperature past the tumour edge.

- P2 Biological effects of cryosurgery — *freeze/thaw cell injury, vascular thrombosis, apoptosis, immunomodulation; argon cools/helium warms; kidney dies at −19.4 °C; <−20 °C only 3 mm inside iceball edge*

Not in the paper answer:

- A 10-min freeze is a compromise: 5 min under-treats, 15 min fractures tissue.
- Thaw is often passive between cycles and active (helium) at the end, so bleeding is handled sooner.
- Probes sit **1–2 cm apart** and come out only after full thaw; early removal fractures tumour and bleeds.
- Heat and cold both give **central** necrosis and a **peripheral sublethal** zone (apoptosis, ischaemia, immunity).
- Cryo target is **≤−40 °C**; the iceball extends **5–10 mm beyond tumour**, pushing the −40 °C isotherm past the tumour edge.
- Heat: **≥60 °C** causes coagulative necrosis; ≥100 °C causes vaporisation and charring. Lethal: 45 °C for 60 min, 55 °C for 5 min, **70 °C for 1 min**.
- **Heat sink:** large vessels carry heat or cold away, so perivascular tumour survives. Counter with colder or longer freezes, more probes, or MWA.

4 · Choice of energy

Why this matters: energies differ in physics, visibility and collateral damage.

RFA electrode	Feature	Strength
Single monopolar	Uninsulated tip	Tumours ≤2 cm only
12-tine umbrella	Tines act as antennas	Most spherical zone
9-tine starburst	Thermistors in tines	Best reproducibility
Cool-tip	Saline-cooled 17 G electrode	Largest zone

	Cryoablation	RFA	MWA
Physics	Argon cools (**Joule–Thomson**); helium warms	450–1200 kHz current causes ionic friction	915 MHz/2.45 GHz causes **dielectric hysteresis**
Target	≤−40 °C	≥70 °C; set 90–105 °C	Often >100 °C
Cycles	**Double freeze–thaw** (10 + 6–8 min)	Two, 30-s cool-down	Single dose (e.g. 65 W × 5 min)
Margin	Iceball 5–10 mm	≥5 mm	≥5 mm
Charring limits	No	**Yes**	No
Grounding pad	No	Yes	No
Heat sink	Yes	Yes	**Least**
Zone visible	**Iceball**	**No**; inferred from impedance and temperature	Gas bubbles on ultrasound (US)
Pros	Spares collecting system	Longest follow-up	Fast; fewest complications
Cons	Bleeding, tumour fracture	Heat sink, pad burns	Risk to adjacent organs

KEY POINT

Why RFA is self-limiting: charring raises impedance and halts current (cap 105 °C, electrodes cooled). MWA *radiates* energy, so it works through desiccated tissue and blood, heats actively to 2 cm (RFA: millimetres) and outruns the heat sink.

- In CA the second freeze enlarges necrosis. RFA impedance protocol: 40–80 W, adding 10 W/min to 130–200 W until impedance is 200–500 Ω. MWA for cT1: 45–65 W for 5–15 min.

5 · Technique

Why this matters: safe probe placement and organ protection prevent most complications.

- **Percutaneous is standard.** Laparoscopic is equally effective but more morbid; keep it for prior scarring or failed hydrodisplacement. General anaesthesia is preferred, as jet ventilation reduces renal motion.

ALGORITHM

1. Give intravenous (IV) antibiotics.
2. Position prone or flank on the computed tomography (CT) gantry.
3. Place probes under US, CT or magnetic resonance imaging (MRI) guidance; confirm on non-contrast CT.
4. Ablate, then contrast CT for completeness and haematoma.

- **Hydrodisplacement** pushes organs away (colon >1 cm) with saline or 5% dextrose (**D5W for RFA**, †as saline conducts current). Caution in cardiac failure.
- **Pyeloperfusion** (central tumours): a ureteral catheter is irrigated at the **opposite temperature**, cold for RFA/MWA and warm for CA.
- **CA:** bleeding rises with size and probe number (>4 cm). **RFA:** ≥70 °C in each quadrant during cool-down excludes heat sink; overlap ablations for larger lesions.
- **MWA:** antennas go tangentially, never into the collecting system: 1 for ≤2.5 cm, 2 for 2.5–3.5, 3 for 3.5–4.5 cm.

6 · Follow-up, success & salvage

Why this matters: with no specimen, imaging alone shows success.

- **Success is complete loss of contrast enhancement** on CT/MRI; the imaging margin replaces the pathological one.
- **Incomplete ablation** enhances on the *first* scan and is re-ablated; same-session imaging should be standard. **Local recurrence** *later* enlarges or enhances and needs biopsy. **Enlargement after any modality is recurrence until proved otherwise.**
- CA zones shrink >50% in year 1; RFA zones barely shrink (**fibrotic halo**); MWA zones are low-attenuation and keep contracting.
- **AUA:** CT/MRI with and without contrast plus chest imaging at **6, 12, 24, 36, 48, 60 months**, then shared decision. US is not routine unless a contrast-enhanced (CEUS) protocol exists.
- Salvage: re-ablation, PN, radical nephrectomy (RN), or stereotactic body radiotherapy (SBRT) if comorbid.

7 · Oncological outcomes vs PN

Why this matters: "as good as surgery?" depends on how recurrence is counted.

- There is no randomised trial. Local recurrence is commoner after TA but **equal once re-ablation is counted**.

- **Metastasis-free and cancer-specific survival (CSS) are equal**; overall survival is lower as patients are older and comorbid.
- **Size predicts failure:** RFA does worse at ≥3 cm, T1b worse than T1a. No energy is superior.

> **KEY POINT**
> **Why comparisons are biased:** benign and under-graded tumours flatter TA; occult pT3a penalises it; non-surgical candidates die of other causes first. The European Organisation for Research and Treatment of Cancer (EORTC) 30904 trial (PN vs RN, ≤5 cm) had only 12 RCC deaths at 9.3 years, so trials need huge cohorts.

8 · Complications & management

Why this matters: complications follow the probe path and thermal field, so prevention is planning.

- Major complications (Clavien ≥3) are **≤5%**, fewer than PN; percutaneous beats laparoscopic, and MWA is lowest (network meta-analysis).

Complication	Prevention	Management
Haemorrhage – commonest; transfusion 2–3%	Full thaw before CA probe removal; avoid intercostal arteries	Start ablating, since RFA/MWA coagulate; if bleeding persists, selective embolisation
Urinoma, fistula, pelviureteric junction (PUJ) or ureteric stricture	Stent and pyeloperfusion; CA for central tumours; tangential antennas	Conservative or stent; percutaneous drain for a large urinoma
Bowel or other organ injury (tumour <1 cm away)	Hydrodisplacement, positioning	Colonephric fistula: stent and low-residue diet; diversion if persistent
Pneumothorax (access above 12th rib)	Recognise during procedure	Aspirate; chest tube rarely
Nerve injury / RFA pad burn	Pads level on posterior thighs	Neuralgia is self-limiting
Infection <1%, may be lethal; abscess at 1–6 months	Urine culture, prophylaxis	†Drain and antibiotics

9 · Newer modalities (investigational)

Why this matters: viva favourites, but none is standard care.

- **High-intensity focused ultrasound (HIFU)** is extracorporeal: heat plus cavitation above 3500 W/cm². Ablation is incomplete (breathing, acoustic interference).
- **Histotripsy** is mechanical (non-thermal) focused ultrasound; no clinical data yet (HOPE4KIDNEY).
- **SBRT/SABR** needs no anaesthesia, but lacks an imaging endpoint, leaves viable tumour on biopsy and has lower CSS; possible salvage role.
- **Irreversible electroporation (IRE)** is non-thermal, opening membrane nanopores; it spares matrix and vessels but controls cancer poorly.

144 Treatment of Advanced Renal Cell Carcinoma

1 · Overview

Why this matters: in metastatic renal cell carcinoma (RCC), drug choice rests on risk group and mechanism.

- P2 Metastatic RCC (mRCC): assessment and management, including recent concepts — *work-up, MSKCC/IMDC, first-line by risk, switch-mechanism later lines, CARMENA/SURTIME, metastasectomy 35–50% 5-yr, KEYNOTE-564, nccRCC*

Not in the paper answer:

- After cytoreductive nephrectomy (CN), only about half in the high-dose interleukin-2 (IL-2) series reached systemic therapy.
- CN is unlikely to help with poor performance status (PS), comorbidity, rapid progression, brain metastases or poor risk.
- CARMENA: 17% of the sunitinib arm later had CN; per-protocol noninferiority was lost.
- Every first-line approval since 2018 is a checkpoint combination tested against sunitinib.
- Low-dose IL-2 gave fewer durable complete responses in two randomised trials.
- Cabozantinib works after VEGF failure because MET and AXL are HIF-driven escape routes.
- About **1/3** present with metastases; a further **20–40%** of localised tumours metastasise later. mRCC 5-yr survival is about **12%**; 10-yr <5%.
- Eras: cytokines, then targeted therapy, now **immune checkpoint (immuno-oncology, IO) combinations**. Adjuvant therapy: *Ch 137*.

2 · Prognostic models

Why this matters: risk group predicts survival and picks first-line therapy.

- **Memorial Sloan-Kettering (MSKCC, Motzer)**: cytokine era, 670 patients on interferon-α (IFN-α). **International Metastatic RCC Database Consortium (IMDC, Heng)**: first-line vascular endothelial growth factor (VEGF)-targeted therapy; validated in 849.

Factor	MSKCC (5)	IMDC (6)
Performance status	Karnofsky <80%	Karnofsky <80%
Haemoglobin	< lower limit of normal (LLN)	< LLN
Corrected calcium	**>10 mg/dL**	> upper limit of normal (ULN)
Lactate dehydrogenase (LDH)	**>1.5 × ULN**	—
Time factor	No prior nephrectomy	**<1 yr** from diagnosis to systemic therapy
Neutrophils; platelets	—	> ULN (each a factor)
Favourable (0) – median overall survival (OS)	20 mo	43.2 mo
Intermediate (1–2)	10 mo	22.5 mo
Poor (≥3)	4 mo	7.8 mo

KEY POINT

Both score host reserve, tumour systemic effects and pace of disease; three factors shared. IMDC swaps LDH and prior nephrectomy for neutrophilia, thrombocytosis and short interval. The IMDC risk group **decides the first-line regimen**.

3 · Biology and drug targets

Why this matters: each drug blocks one step of the VHL pathway or one immune brake.

- The **von Hippel–Lindau (VHL)** protein (gene on 3p), with elongins B/C and cullin-2 (CUL2), tags hydroxylated **hypoxia-inducible factor (HIF)-α** for degradation.
- When VHL is lost, HIF drives VEGF, platelet-derived growth factor (PDGF), TGF-α, GLUT-1 and erythropoietin. Drugs block the ligand (bevacizumab), receptors (tyrosine kinase inhibitors, TKIs), HIF translation (mammalian target of rapamycin, mTOR) or HIF-2α (belzutifan). MET/AXL/FGFR escape explains broader later TKIs.
- **CTLA-4** and **PD-1/PD-L1** switch off T cells. Their inhibitors release the brake, so toxicity is autoimmune and benefit shows as OS and durability rather than progression-free survival (PFS).
- VEGF inhibitors are immunomodulatory, the rationale for IO plus TKI. Clear cell RCC (ccRCC) is chemoresistant (response 5.5–6%).

Agent	Target / route	Signature toxicity / note
Bevacizumab	Antibody vs VEGF-A; IV	Bleeding, proteinuria, hypertension (HTN)
Sorafenib	VEGFR2, PDGFR-β, raf-1	Rash, hand–foot syndrome (HFS); rarely used
Sunitinib	VEGFR2, PDGFR-β, c-KIT, FLT3; **50 mg, 4 wk on / 2 off**	Diarrhoea, HFS, fatigue, HTN, thrombocytopenia, **hypothyroidism**
Pazopanib	VEGFR, PDGFR, KIT	**Hepatotoxicity (raised alanine aminotransferase)**; less fatigue/HFS
Axitinib	VEGFR1–3 only – most selective	Diarrhoea, fatigue, HTN
Tivozanib	VEGFR1–3, c-KIT, PDGFR-β	Third line
Cabozantinib	VEGFR, **MET, AXL**	Fatigue, diarrhoea, HFS; frequent dose reduction
Lenvatinib	VEGFR1–3, **FGFR1–4**, PDGFR-α, RET, KIT	Partner of everolimus or pembrolizumab
Temsirolimus	mTOR; **IV weekly**	Mucositis, rash, high glucose, low phosphate, high cholesterol, pneumonitis
Everolimus	mTOR; oral daily	Asthenia, anaemia
Belzutifan	**HIF-2α inhibitor**; oral	After IO and TKI; also VHL-associated RCC
Nivolumab, pembrolizumab	Anti-PD-1; IV	Immune-related: colitis, hypophysitis, vitiligo
Avelumab	**Anti-PD-L1**; IV	
Ipilimumab	Anti-CTLA-4; IV	

4 · Surgery in metastatic disease

Why this matters: surgery now serves systemic therapy; the question is who and when.

- **CN rationale:** spontaneous regression of metastases is <1%, and a bulky primary suppresses T cells. CN helps only as a platform for systemic therapy.
- Fewer than 1/4 reach systemic therapy if they have Eastern Cooperative Oncology Group (ECOG) PS 2, multiple sites, or brain or liver metastases.
- **CARMENA caveats:** about 45% were poor risk; about 25% of the CN arm were off-protocol; it stopped early and was underpowered; and sunitinib is no longer standard. It does **not** show harm in fit, low-burden, favourable patients.

ALGORITHM

1. Assess mRCC: histology, IMDC risk, PS and metastatic burden.
2. If fit, with low burden, favourable features and a resectable primary, then consider upfront CN followed by systemic therapy.
3. If intermediate/poor risk or high burden, then give systemic therapy first and offer deferred CN to responders.

KEY POINT

CN helps only patients fit enough, and with disease slow enough, to reach effective systemic therapy; operating first on poor-risk patients mainly delays it.

- **Metastasectomy** does best with complete resection, a solitary lesion, age <60, disease-free interval >1 yr, lung site, size <4 cm and metachronous disease. All are markers of indolent biology.
- **Palliative** nephrectomy treats pain, haematuria and paraneoplastic syndromes, but try **angioembolisation for haematuria** first; only about half of hypercalcaemic patients improve.
- Brain, weight-bearing and spinal metastases get surgery and radiation.

5 · Systemic therapy for ccRCC

Why this matters: first line follows IMDC risk; later lines switch mechanism.

	IFN-α	High-dose IL-2
Response	10–15%	15–20%
Durable complete response (CR)	<2%	**7–9%**; >60% of CRs stay disease-free
Regimen	Schedule/route immaterial	**600,000–720,000 IU/kg** IV bolus 8-hourly, max 15 doses
Toxicity	Constitutional	**Vascular leak**: hypotension, multiorgan failure; early mortality 2–5%
Current role	Not used first-line	Carefully selected fit patients

- IL-2 is the only agent approved on phase II response. Best candidates have clear cell histology, good PS and prior nephrectomy.

Risk	Preferred / standard	Alternatives (incl. if IO not possible)	Certain circumstances
Favourable	Pembrolizumab + axitinib; nivolumab + cabozantinib; pembrolizumab + lenvatinib	Avelumab + axitinib, cabozantinib, ipilimumab + nivolumab, pazopanib, sunitinib	**Active surveillance**, axitinib, high-dose IL-2

Risk	Preferred / standard	Alternatives (incl. if IO not possible)	Certain circumstances
Intermediate / poor	Same three **+ ipilimumab + nivolumab**	Cabozantinib, avelumab + axitinib, pazopanib, sunitinib	Axitinib, high-dose IL-2, **temsirolimus**

KEY POINT

Choosing: IO/TKI gives higher response and PFS and is better tolerated; IO/IO gives deeper durable responses (CR 9%) with autoimmune toxicity; they have never been compared directly. Ipilimumab + nivolumab is restricted to intermediate/poor risk (favourable did better on sunitinib in CheckMate 214). Avelumab + axitinib is not NCCN-preferred (no OS benefit). PD-L1 does not select. Use a TKI or temsirolimus if IO is contraindicated.

Prior therapy	European Association of Urology (EAU) standard	National Comprehensive Cancer Network (NCCN) recommended (no preferred)
TKI only (IO-naive)	**Nivolumab or cabozantinib** (alternative axitinib)	IO combinations, nivolumab, cabozantinib, lenvatinib + everolimus
Prior IO	Any VEGF-targeted agent not previously used	Axitinib, **belzutifan**, cabozantinib, lenvatinib + everolimus, **tivozanib**

- Tivozanib is used after ≥2 prior therapies.
- Failed combinations: sunitinib + mTOR and nivolumab + sunitinib/pazopanib were too toxic; bevacizumab + mTOR gave no gain (RECORD-2, INTORACT).
- **Ipilimumab + nivolumab toxicity:** grade 3–4 events **46%**, high-dose steroids 35%, discontinuation 22%. The dose (nivolumab 3 mg/kg + ipilimumab 1 mg/kg 3-weekly × 4) came from CheckMate 016.
- Nivolumab alone vs everolimus, grade 3–4: 19% vs 37%. Autoimmune events with ipilimumab predicted response.
- † For immune-related events, hold IO and give steroids for grade ≥2 (prednisolone 1–2 mg/kg for grade 3–4), tapering over ≥4 weeks. Grade 4 means permanent stopping.
- TKI toxicity is managed by dose reduction or interruption; avoid pazopanib if hepatotoxicity risk. With mTOR inhibitors, monitor glucose, lipids, phosphate and pneumonitis.

6 · Non-clear cell RCC

Why this matters: ccRCC drugs work poorly here; some tumours have a targetable driver.

- Non-clear cell RCC (nccRCC) is 15–25%. No agent is proven, so offer a **trial to all**.
- Papillary: sunitinib response is 5–10%. Temsirolimus beat IFN in the nccRCC subgroup. Sunitinib vs everolimus showed a trend to sunitinib. Chromophobe: trend to everolimus.
- **Hereditary papillary RCC (HPRC)** carries a germline activating **MET** mutation (chromosome 7) and gives type 1 papillary RCC. Sporadic papillary has **trisomy 7 in >2/3** and somatic MET in about 13%. Foretinib, savolitinib and crizotinib are active in MET-driven tumours.
- **Hereditary leiomyomatosis and RCC (HLRCC)**: fumarate hydratase loss gives type 2 papillary RCC. The cells depend on glycolysis, so **bevacizumab + erlotinib** restricts glucose (partial response about 2/3).
- **Collecting duct** carcinoma is treated with **gemcitabine + carboplatin**. **Sarcomatoid** tumours: gemcitabine-based therapy; they have more PD-1-positive infiltrates.

KEY POINT

In nccRCC the syndrome names the driver, and the driver names the drug: VHL leads to VEGF/mTOR/HIF-2α drugs, MET to MET inhibitors, and FH loss to glucose restriction.

7 · Landmark trials

Why this matters: each line of therapy rests on these trials.

Trial	Design	Result	Take-home
SWOG 8949 / EORTC	CN + IFN vs IFN	OS 11.1 vs 8.1; 17 vs 7 mo; response same	CN in cytokine era
CARMENA	CN + sunitinib vs sunitinib; noninferiority; n=450	OS 13.9 vs **18.4** mo, hazard ratio (HR) 0.89	No upfront CN in intermediate/poor
SURTIME	Immediate vs deferred CN	OS 15.0 vs **32.4** mo, HR 0.57	Systemic therapy first is safe
Sunitinib vs IFN	Phase III, first line	PFS 11 vs 5 mo; response 31 vs 6%	Later comparator
COMPARZ	Pazopanib vs sunitinib; noninferiority	PFS HR 1.05; OS similar	Equivalent; quality of life and preference (PISCES) favoured pazopanib
CALGB 90206 / AVOREN	Bevacizumab + IFN vs IFN	PFS better, OS not	Little used
Temsirolimus phase III	vs IFN vs combination; poor risk, all histologies	OS 10.9 vs 7.3 mo	Only mTOR inhibitor with OS gain
TARGET	Sorafenib vs placebo after cytokines	PFS 5.5 vs 2.8 mo; OS same	Historical
Everolimus phase III	vs placebo after sunitinib/sorafenib	PFS 4.0 vs 1.9 mo; OS same	Supplanted as monotherapy
Lenvatinib ± everolimus	Phase II, after VEGF therapy	Combination PFS longer	Lenvatinib + everolimus
AXIS	Axitinib vs sorafenib, second line	PFS 6.7 vs 4.7; OS same	Second-line axitinib
METEOR	Cabozantinib vs everolimus after VEGF	PFS HR 0.58; **OS HR 0.70**	Only TKI with OS gain over active comparator
CABOSUN	Phase II; cabozantinib vs sunitinib; intermediate/poor	PFS 8.2 vs 5.6, HR 0.66	First-line cabozantinib
TIVO-3	Tivozanib vs sorafenib, after 2–3 lines	PFS HR 0.73; OS same	Third-line tivozanib
LITESPARK-005	Belzutifan vs everolimus after IO + TKI	PFS HR 0.75 (curves separate late)	HIF-2α inhibition approved
CheckMate 025	Nivolumab vs everolimus after TKI; primary OS	OS 25.0 vs 19.6, **HR 0.73**; PFS same	IO benefit shows as OS
CheckMate 214	Ipilimumab + nivolumab vs sunitinib	Intermediate/poor OS HR 0.63; CR 9 vs 1%	Only IO/IO; not for favourable
KEYNOTE-426	Pembrolizumab + axitinib vs sunitinib	PFS and OS better; response 60 vs 40%	All risk groups

Trial	Design	Result	Take-home
JAVELIN Renal 101	Avelumab + axitinib vs sunitinib	PFS better; OS not significant	Not NCCN-preferred
CheckMate 9ER	Nivolumab + cabozantinib vs sunitinib	PFS 16.6 vs 8.3, HR 0.51; OS HR 0.70	All risk groups
CLEAR	Pembrolizumab + lenvatinib vs sunitinib	PFS 23.9 vs 9.2, **HR 0.39**; response 71%	Largest PFS/response effect

★ Putting it together

A · Small renal mass

Four options, chosen by fitness, size and renal function.

	Active surveillance (AS)	Ablation (Ch 143)	Partial nephrectomy (PN)	Radical nephrectomy (RN)
Best for	Frail, life expectancy <5 y; solid <2 cm or mainly cystic	cT1a **≤3 cm**, comorbid; biopsy first	cT1a standard; imperative if solitary, bilateral, hereditary, chronic kidney disease (CKD)	Only if high complexity, no CKD/proteinuria, and new eGFR **>45**
Numbers	Growth 0.1–0.3 cm/y; metastasis <2%; treat if >3 cm or >5 mm/y	Cryo: 2 freeze–thaw cycles; RFA/MWA >60–70 °C	Trifecta: R0, no complication, warm ischaemia ≤25 min; margin width immaterial	New GFR ≈ GFR × contralateral split × 1.25
Weakness	Imaging adherence	More local recurrence (re-ablate); no pathology	Leak, bleeding, pseudoaneurysm	CKD 65% vs 20% after PN

B · Renal lesions

Imaging and markers sort benign from malignant before anyone operates.

Lesion	Imaging	Genetics / IHC	Action
Simple cyst	≤20 HU, no enhancement	—	Nothing
Complex cyst	Bosniak v2019	—	≤2 cm surveil; >4 cm excise
Angiomyolipoma (AML)	Fat ≤−20 HU; fat-poor = trap	TSC2/TSC1; HMB-45, Melan-A+	Embolise if >6 cm, aneurysm >5 mm or bleeding; everolimus in tuberous sclerosis
Oncocytoma	Central scar, enhancement inversion	KIT+, CK7 single cells; Birt–Hogg–Dubé	Cannot exclude chromophobe, so surveil or PN
Clear cell RCC	Hypervascular/heterogeneous	3p/VHL loss activates HIF-2α; CA-IX	PN/RN/ablation/AS
Papillary RCC	Hypoenhancing	Trisomy 7, 17; loss of Y; MET	As RCC
Chromophobe RCC	Moderate enhancement	Multiple losses; diffuse CK7	As RCC; good prognosis
Lymphoma/metastasis	Infiltrative, hypovascular	—	**Biopsy; don't operate**
Upper tract urothelial carcinoma (UTUC)	Filling defect, ~46 HU, cytology+	MSI suggests Lynch	Risk-stratify, then KSM or RNU

C · UTUC: kidney-sparing (KSM) vs RNU

Risk group decides between saving the kidney and radical nephroureterectomy (RNU).

	KSM (ureteroscopic, percutaneous, segmental)	RNU + bladder cuff
Who	Low risk: ALL of unifocal, <2 cm, low grade, no invasion on CT urogram; or imperative (solitary, CKD, bilateral)	ANY of multifocal, >2 cm, high grade, hydronephrosis, prior cystectomy, variant
Extras	Ureteroscopy at each surveillance; UGN-101 mitomycin gel (low grade); ~3× re-interventions	≤12 wk; template lymph node dissection; single post-op intravesical chemo
Systemic	—	Neoadjuvant platinum if post-op CrCl <50; adjuvant (POUT) ≥pT2/N+

D · Open vs minimally invasive

Minimally invasive is the default; open is kept for difficult cases.

Open (Ch 141)	Laparoscopic / robotic (Ch 142)
Locally advanced (IVC thrombus needing bypass); cytoreduction; complex anatomy; prior surgery; trauma (midline, early vascular control)	Standard for organ-confined RN and most PN: less pain, blood loss and stay; same oncology; retroperitoneal is shorter

E · Metastatic RCC

Risk group sets first-line therapy; surgery is selective.

ALGORITHM

1. **Score IMDC (6 factors)**: KPS <80%, Hb < LLN, calcium/neutrophils/platelets > ULN, <1 yr to therapy. 0 is favourable, 1–2 intermediate, ≥3 poor.
2. **First line**: IO + TKI (pembro + axitinib, nivo + cabozantinib, pembro + lenvatinib) at any risk; ipi + nivo only if intermediate/poor.
3. **Cytoreductive nephrectomy**: upfront only if fit, low burden, favourable; otherwise deferred in responders (CARMENA, SURTIME).
4. **Later lines**: switch mechanism; belzutifan (HIF-2α) after IO + TKI.